



Energy Audit
for
Four Seasons Hotel, Sydney

November 2020

Revision 0

Enman Pty Ltd

Experienced, Professional People at Your Service

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Executive Summary

Enman Pty Ltd was engaged by Four Seasons Hotel, Sydney to carry out this energy audit under the City of Sydney's Environmental Performance Ratings and Assessment Grants to provide insight into the hotel's energy use and energy saving opportunities including recommendations. This is a level 2 energy audit in accordance with energy audit standard AS/NZS 3598.1-2014 Energy audits - Commercial buildings. The boundary of the audit is the hotel's building services. A site visit was conducted on 22 October 2020.

This report outlines the findings of the energy audit and aims to provide:

- Understanding of the energy use on site
- Benchmarking of the hotel's energy performance
- Identification of energy saving projects with cost-benefit analysis
- Recommendations and an action plan

Benchmarking

A NABERS rating has been carried out to develop a benchmark of the hotel's energy efficiency. The NABERS star rating indicates that the current energy efficiency is average (3.5 stars), and water is average (3 stars). If all recommended projects are implemented, there will be an estimated 12% reduction in electricity consumption and a 2% reduction in gas consumption.

Yearly Energy Use and Tariff

The electricity consumption is 6,674,273kWh/y and the gas consumption is 17,461 GJ/y.

Current Energy Costs:

Peak variable electricity cost, \$/kWh	0.18
Shoulder variable electricity cost, \$/kWh	0.15
Off-Peak variable electricity cost, \$/kWh	0.10
Overall variable electricity cost, \$/kWh	0.13
Demand Cost \$/kVA/annum	119.76
Gas Variable Cost, \$/GJ	11.92

*'Variable' electricity costs include all fees charged on a c/kWh basis, including retailer, network, AEMO and environmental charges. It excludes fixed costs and demand costs.

The current variable cost for both electricity and gas have been used in the savings calculations. The costs of electricity and gas used in the saving calculation are as follows.

Projected Energy Costs:

Cost of electricity is 0.13 \$/kWh

Cost of gas is \$11.92 /GJ

Greenhouse Gas Emissions

The site's greenhouse gas emissions can be calculated as shown in the table below. The total current greenhouse gas emissions are 7,264 tCO₂-e/y.

Table: Site Greenhouse Gas Emission

Energy Source	Consumption (kWh/GJ)	GHG Coefficient	GHG Emissions/y	Total cost \$/y
Electricity	6,674,273	0.92	6,140	\$1,055,902
Natural Gas	17,461	64.33	1,123	\$351,821
TOTAL			7,264	\$1,407,723

*Listed GHG coefficients are the current Scope 2 & 3 emissions factors for NSW (electricity) and the current Scope 1 & 3 emissions factors for natural gas distributed in a pipeline (NSW) from the National Greenhouse Accounts Factors (NGAF) 2018. Factors used by NABERS Rate at present are slightly different (0.96 kgCO₂e/kWh for electricity and 64.3 kgCO₂e/GJ for gas).

Energy Saving Summary

There are a few energy saving opportunities have been identified and preliminary design works for those have been carried out to estimate the project implementation cost.

To estimate the energy saving an energy usage database has been created. The end-use data has been compared with the actual bills. The difference between actual and estimated energy use is minimal and acceptable.

The energy saving opportunities are summarised below.

Energy Saving Summary

Project Name	Electricity Saving (kWh/y)	Gas Saving (GJ/y)	Demand Saving (kVA)	Cost saving (\$/y)	GHG saving (tCO ₂ -e/y)	Project cost (\$)	Simple Payback (y)	Estimated ESC Revenue ¹	Revised Payback with ESC
Demand based Control of AHU Fan Speed	77,789	-	-	10,108	72	24,160	2.4	11,376	1.3
VSD on FCU Fans	22,159	-	-	2,879	20	12,142	4.2	3,232	3.1
Demand Based FCU Fan Speed Control	44,371	-	-	5,766	41	8,000	1.4	6,480	0.3
Car Park Fan Optimal Control	18,336	-	-	2,383	17	4,000	1.7	2,672	0.6
Refrigeration System Upgrade and Heat Recovery	330,739	24	-	43,262	306	253,918	5.9	48,608	4.7
Lighting Upgrade	234,402	-	14	32,135	216	132,040	4.1	34,304	3.0
Air Curtain on Two Main Entrance and Exit Doors	44,774	-	-	5,818	41	20,000	3.4	6,544	2.3
Upgrade of Old Thermal Storage Tanks	597	299	-	3,635	20	160,000	44.0	2,688	43.3
Pressure Booster System Upgrade	23,652	-	-	3,073	22	71,044	23.1	3,456	22.0
Hot Water Pipe Insulation	-	33	-	398	2	1,000	2.5	288	1.8
TOTAL	796,820	356	14	109,456	756	686,304	9.3	119,648	8

¹ ESC revenue is an estimate only

Recommendations

Enman provide recommendations on action plan for project implementation as short-term, medium-term and long-term. These are based on the economic incentive, equipment life, plant reliability and redundancy.

Enman's recommendations for the energy saving projects are as follows:

Short-term implementation projects

- | | |
|---|--------------|
| • Demand based Control of AHU Fan Speed | Refer to 6.1 |
| • Car Park Fan Optimal Control | Refer to 6.3 |
| • Lighting Upgrade | Refer to 6.5 |
| • Air Curtain on Two Main Entrance and Exit Doors | Refer to 6.6 |
| • Hot Water Pipe Insulation | Refer to 6.9 |

Medium-term implementation projects

- | | |
|--|--------------|
| • VSD on FCU Fans | Refer to 6.2 |
| • Refrigeration System Upgrade and Heat Recovery | Refer to 6.4 |

Long-term implementation projects

- | | |
|--|--------------|
| • Upgrade of Old Thermal Storage Tanks | Refer to 6.7 |
| • Pressure Booster System Upgrade | Refer to 6.8 |

1 Introduction

This audit was commissioned by Four Seasons Hotel, Sydney with funding from the City of Sydney's Environmental Performance Ratings and Assessment Grants. This audit has been carried out to identify energy saving opportunities at the hotel. The hotel is located at 199 George St, The Rocks NSW 2000. It has 531 guest rooms, a few function rooms, and restaurants.

A site survey was carried out on 22 October 2020.

The electricity meter (NMI NCCCZ01159) supplies electricity to:

- Chiller
- Cooling Tower
- Pumps
- AHUs and Fans
- Lifts
- Lighting
- Kitchens
- Guest room appliances
- All other equipment in the hotel

There is a gas meter which supplies gas to:

- Boilers for domestic and heating hot water
- Kitchen

Know your auditor:

The audit has been carried out by the following auditors:

Supervisory auditor: Anwar Ahmed

Energy auditor: Rasel Mahamud

2 Historical Energy Consumption

2.1 Electricity Consumption

The hotel electricity consumption from March 2019 to Feb 2020 was 6,695,523 kWh. Consumption varies over months and significantly higher in summer months than winter. This indicates energy consumption for cooling of the building is performed by electricity. Consumption data for the total site is shown in Table 1: Electricity Consumption Monthly Data and Figure 1: Electricity Consumption Daily Average.

Table 1: Electricity Consumption Monthly Data

Month	Total Energy (kWh)	Max Demand, kVA	Total Cost
Mar-19	637,750	1,132	\$102,474
Apr-19	573,273	1,122	\$88,920
May-19	559,166	980	\$89,783
Jun-19	512,421	978	\$83,123
Jul-19	502,653	876	\$82,059
Aug-19	493,289	858	\$80,088
Sep-19	491,083	886	\$76,934
Oct-19	531,456	1,011	\$82,686
Nov-19	544,781	1,041	\$87,032
Dec-19	609,445	1,067	\$94,567
Jan-20	643,942	1,133	\$96,548
Feb-20	596,265	1,107	\$91,688
TOTAL	6,695,523	1,133	\$1,055,902

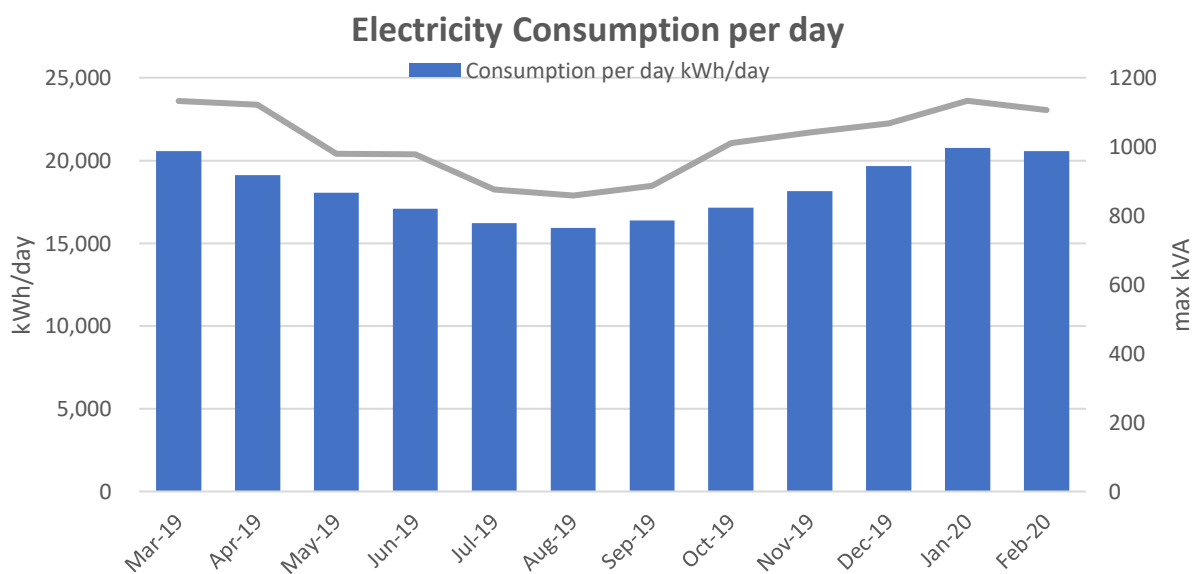


Figure 1: Electricity Consumption Daily Average

2.2 Gas Consumption

The hotel gas consumption from Mar 1, 2019 to Feb 29, 2020 was 17,461 GJ which varies over months. Consumption data is shown in Table 2: Gas Consumption Monthly Data and Figure 2: Gas Consumption Daily Average. The gas consumption increases in the winter months which indicate building heating is performed by gas.

Table 2: Gas Consumption Monthly Data

Period	Total Energy (GJ)	Total Cost (\$)
Feb 12-Mar 11	1,207	\$29,430
Mar 12-Apr 9	1,304	\$30,758
Apr 10-May 10	1,679	\$36,795
May 11-Jun 12	2,013	\$41,832
Jun 13-Jul 11	1,904	\$40,439
Jul 12-Aug 12	1,939	\$31,852
Aug 13-Sep 11	1,554	\$27,149
Sep 12-Oct 14	1,352	\$24,976
Oct 15-Nov 13	1,148	\$22,393
Nov 14-Dec 13	1,158	\$22,700
Dec 14-Jan 14	1,115	\$22,464
Jan 15-Feb 13	1,089	\$21,784
TOTAL	17,461	\$352,572

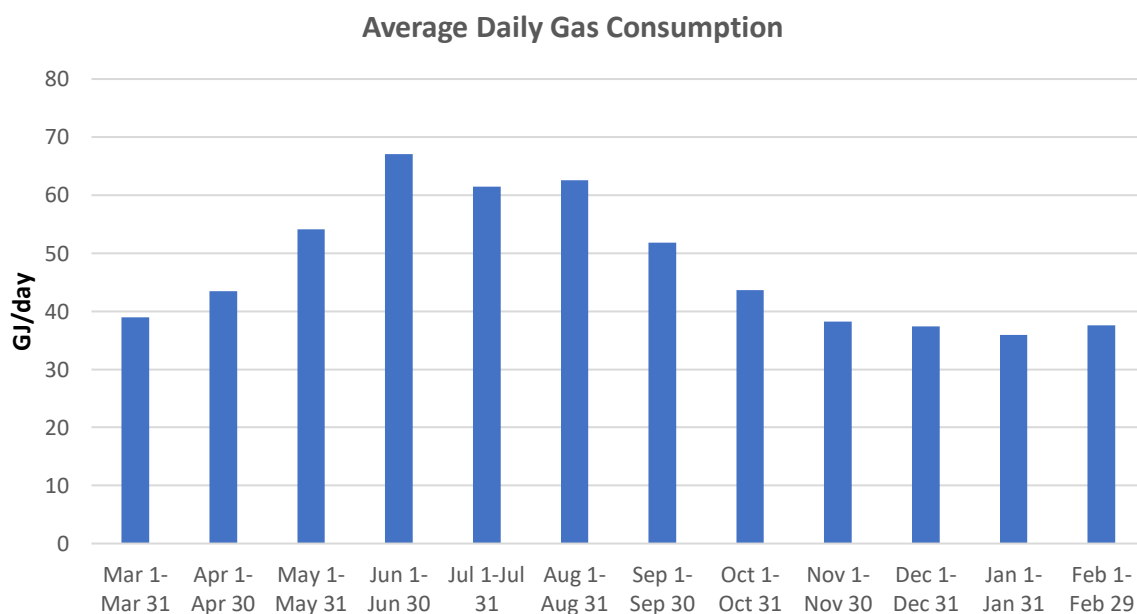


Figure 2: Gas Consumption Daily Average

2.3 NABERS Rating

The National Australian Built Environment Rating System (NABERS) was used to rate the energy and water consumption of the hotel against other similar hotels, considering factors such as the climate and facilities it provides. This gave a rating of 3.5 stars for energy and 3 stars for water based on Mar 2019 – Feb 2020 bills. The rating provides valuable information about the relative consumption of the hotel. The 3.5 stars represents an over average and 3 stars an average according to the NABERS scale shown below.

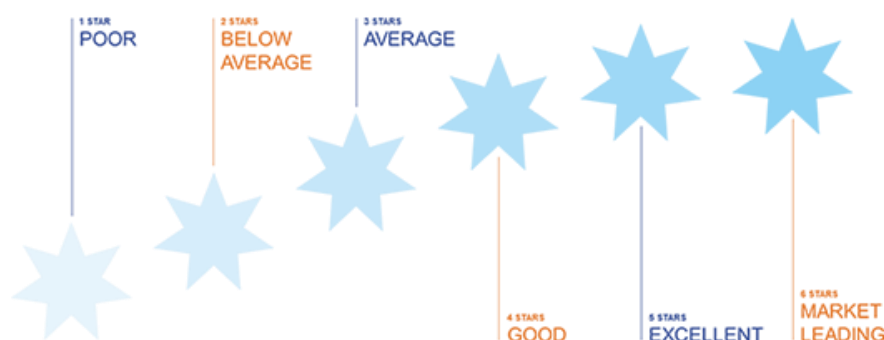


Figure 3: The NABERS Rating Scale

Table 3: NABERS Results shows the input and output data for the NABERS ratings.

Table 3: NABERS Results

Premises	
Premises Type:	Hotel
Postcode:	2000
Guest Rooms:	498.5
Hotel Star Rating:	5 stars
Laundry Serviced Rooms:	Nil
Surface Area of Heated Pools:	117.5
(NABERS Rated) Seating Capacity in Function Rooms:	1723.5 seats
Rating Period:	01/03/2019 to 29/02/2020
Energy Consumption	
Electricity:	6,695,523 kWh
Gas:	17,461,288 MJ
Energy Rating	
NABERS Energy Rating for the hotel	3.5 Stars ★★☆☆☆
Total greenhouse gas emissions	7,146,985 kgCO ₂ -e p.a.
Greenhouse gas emissions per room	14,337 kgCO ₂ -e/room p.a.
Average greenhouse gas emissions per room for a comparable hotel ² with a 2.5 stars	21,498 kgCO ₂ -e/room p.a.

² This value indicates the greenhouse gas emission per room for a comparable hotel with 2.5 stars

Water Rating	
NABERS Rating	3 Stars      
Actual water consumption	77,123 kL
Water consumption per room	154.71 kL/room p.a.
Average water consumption per room for a comparable hotel ³ with a 2.5 stars	179.45 kL/room p.a.

The results of energy and water rating demonstrate that both energy and water consumption of the hotel falls within the average band of similar hotels. Therefore, the hotel has opportunities to improve both of its energy and water ratings by reducing consumptions to be a good or an excellent performer.

2.4 End-Use Breakdown

The energy end-use breakdown is an estimation of the consumption of each end-use which is compiled from a list of equipment with kW ratings and estimations of run time and loading factors. The full breakdown database is provided in Appendix B: Energy End-Use Breakdown.

Electricity

The energy end-use breakdown for electricity is summarised in Table 4 and Figure 4.

Table 4: Electricity End-Use Breakdown

Description	Energy Consumption (kWh/y)	%
Chiller	777,693	11.6%
CT	26,434	0.4%
AHU	555,822	8.3%
FCU	1,160,968	17.3%
Fan	356,769	5.3%
Pump	333,280	5.0%
ACCU	145,171	2.2%
Lift	405,238	6.1%
Refrigeration	57,974	0.9%
Water Heater	424,159	6.3%
Air Compressor	33,726	0.5%
Kitchen Appliance	334,776	5.0%
Light	709,733	10.6%
Room Appliances	292,676	4.4%
Miscellaneous	1,081,104	16.1%
Total	6,695,523	100%

³ This value indicates the water consumption per room for a comparable hotel with 2.5 stars

ENERGY CONSUMPTION KWH/Y

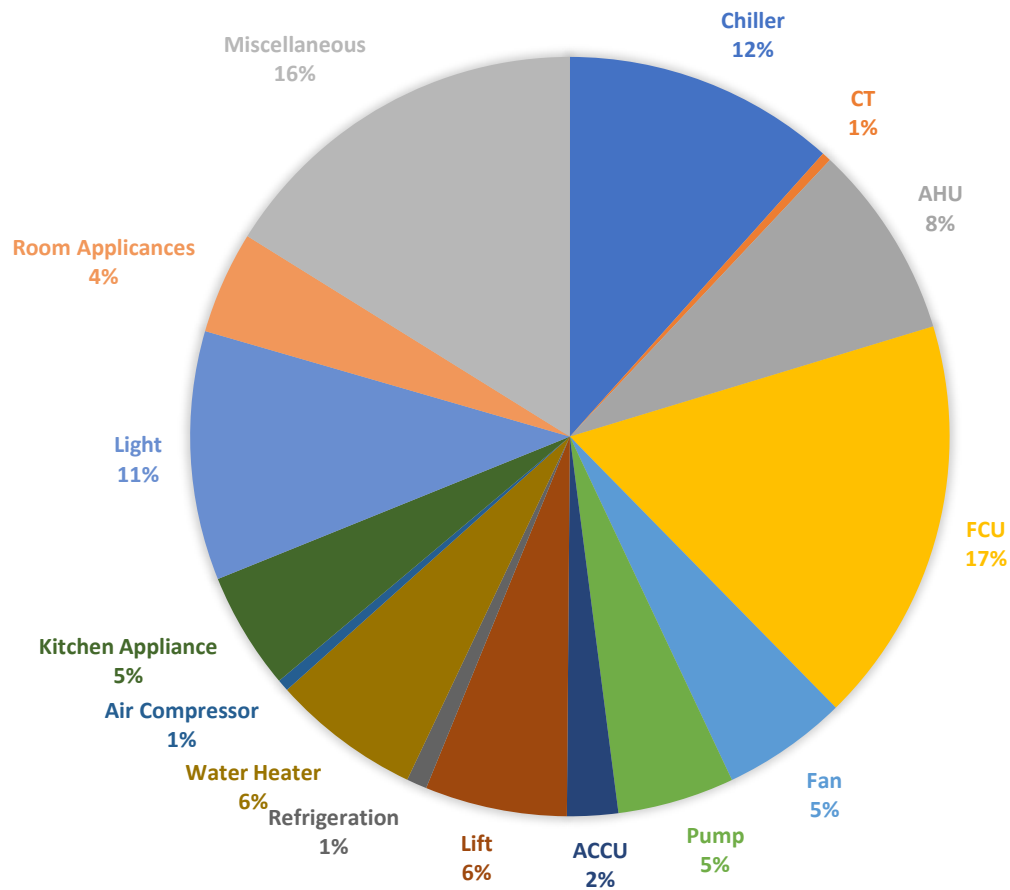


Figure 4: Electricity End-Use Breakdown Chart

The pie chart shows electricity consumption from the main end uses. The largest end-user is the HVAC which includes chiller, CT, Pumps, AHUs and FCUs. Together they consume nearly half of hotel's bill. Lighting has considerable contribution of 11% to the electricity bill of the hotel.

Gas

The energy end-use breakdown for gas is shown in Table 5 and Figure 5 below.

Table 5: Gas End-Use Breakdown

Description	Gas Consumption (GJ/y)	%
Hot Water Boiler	11,941	69%
Steam Boiler	3,449	20%
Kitchen, Others	2,033	12%
Total	17,424	

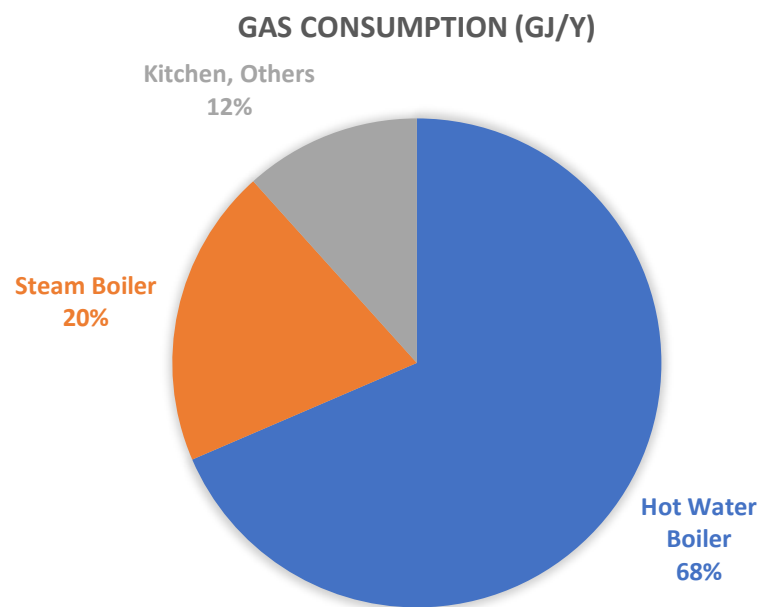


Figure 5: Gas Consumption Distribution

The pie chart shows gas consumption. The hot water boilers consume nearly 70% of the hotel's gas and is the biggest user.

2.5 Energy Profile and Demand

The electrical demand profile of the hotel is shown in the figures below for typical days in summer and winter, as well as for the peak demand day in the year 2019. The average power factor at the site 0.98 which is good. It goes as low as 0.95.

In the summer month, the demand drops to below 800 kW overnight and increases again from early morning (around 6:00 am) to a maximum of 1100 kW depending on the temperature of the day. The power starts reducing again from around 6 pm. The load profile of summer days is averaged and presented in Figure 6: Summer Power Profile.

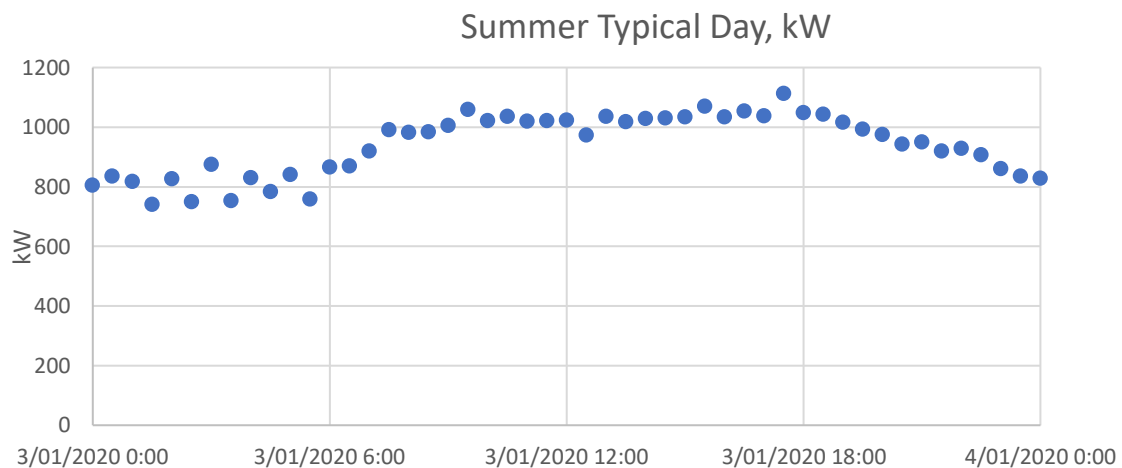


Figure 6: Summer Power Profile

The peak demand for the year 2019 occurred on 22nd January at 9:00 am. It was like a typical summer day of the year, but the excessive demand can be triggered by simultaneous start-ups of several large energy-consuming equipment. The demand profile of a few consecutive days before and after the peak is presented in Figure 7: Maximum Demand Day .

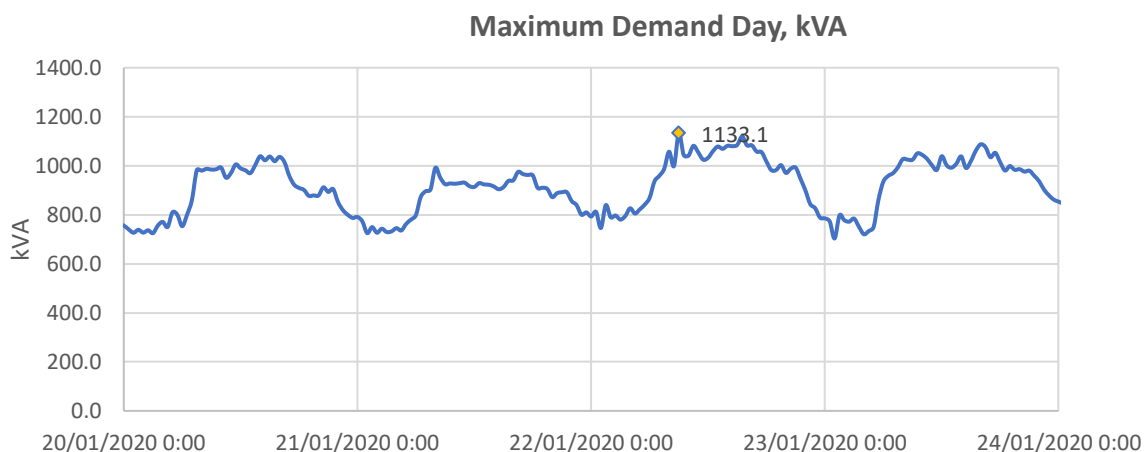


Figure 7: Maximum Demand Day

The load profile of a winter day is shown in Figure 8: Winter Power Profile. The load profile is much flatter than the summer due to lower cooling demand during the day.

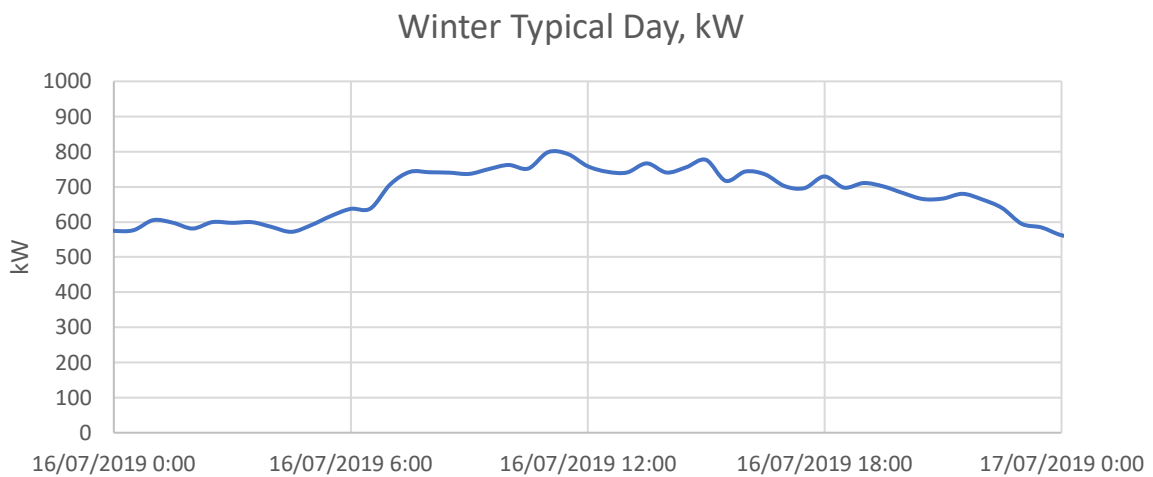


Figure 8: Winter Power Profile

The daily energy consumption and maximum and minimum temperature of the day are compared in Figure 9. There is a close relationship between temperatures and energy consumption. When the temperature goes very high there is high electricity consumption due to building cooling.

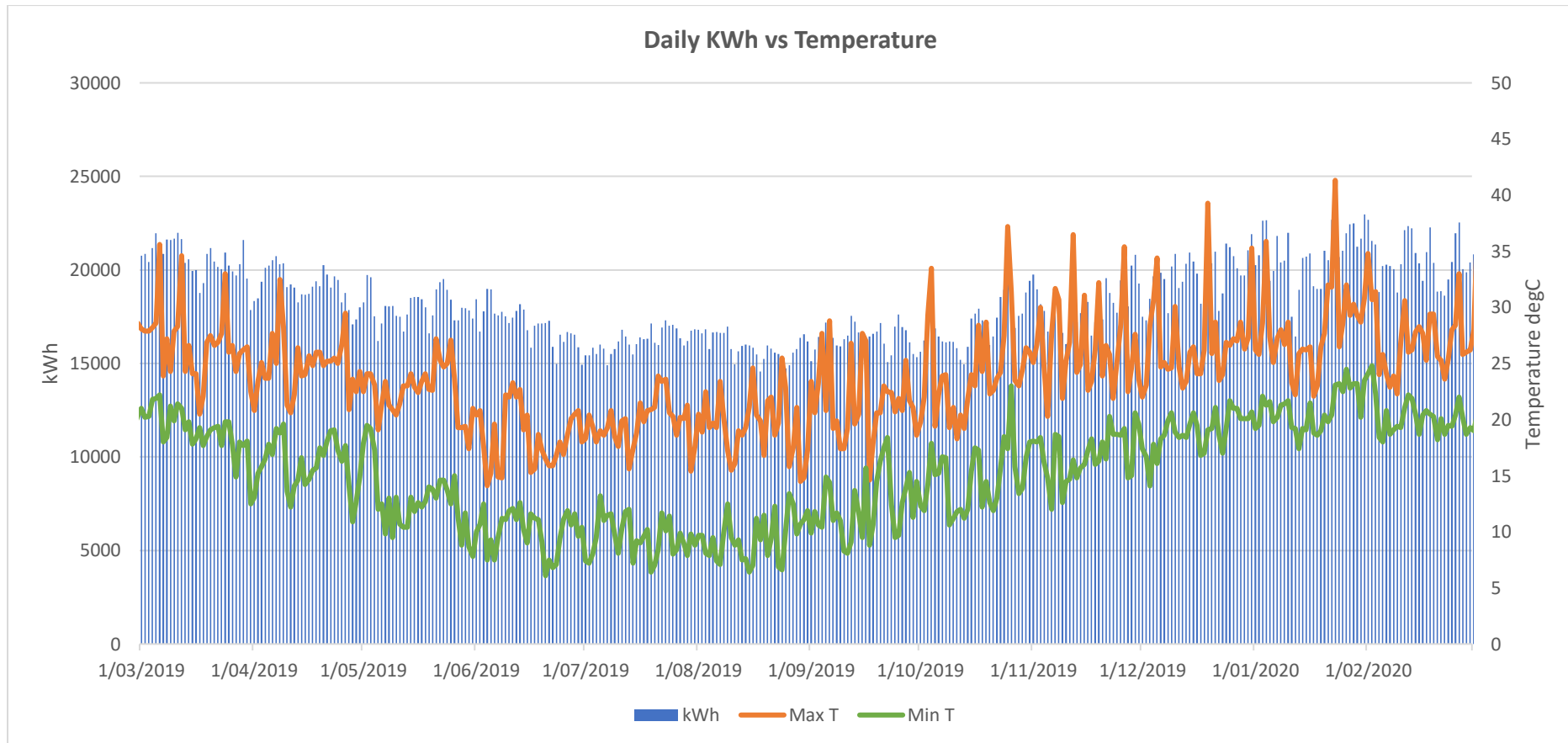


Figure 9: Comparison of Energy Consumption and Temperature

3 Energy Management Practices

David Perrett, Director of Engineering is responsible for energy management at the hotel. There is no written energy policy for the hotel as part of its energy management. However, the hotel engineering department provides great attention to reduce energy consumption. They have limited submetering of electrical energy which can be tracked through BMS.

Approval for energy saving projects or investment decision is based on energy savings, asset life and building health/performance for comfort.

The hotel building was built in 1982 and it has renovated in last year. There are many original equipment are still running. The engineering office performs both preventative and reactive maintenance to keep the hotel asset safe and operational. The maintenance system is supplied by Simpro.

The hotel has a Building Management System (BMS) for building cooling, heating and air-conditioning. The BMS is supplied by ENMAN/ENGIE and has some energy management functionality. The hotel has a room management system. The hotel guest room setpoint temperatures are altered both winter and summer when unoccupied to save energy.

4 Tariffs

4.1 Electricity

Electricity for the hotel is currently supplied by Origin through the Ausgrid distribution network. The applicable charges are shown in Table 6 below. The average variable electricity cost using the rates of utility provider and current network charges is **21 c/kWh** (16% peak, 33% shoulder and 51% off-peak plus other charges).

Table 6: Electricity Rates – February 2020

Retailer Energy Charges	
Peak	11.209 c/kWh
Shoulder	11.209 c/kWh
Off-Peak	7.577 c/kWh
Network Charges	
Network Peak	4.6546 c/kWh
Network Shoulder	1.7939 c/kWh
Network Off Peak	0.8582 c/kWh
Capacity Charge	32.811 c/kVA/Day
Access Charge	24.949242 \$/Day
Other Charges	
Total Environmental Charges	1.82795299 c/kWh
Total Market Fee	0.076881 c/kWh

4.2 Gas

Gas for the hotel is supplied by ORIGIN through the Jemena Distribution Zone. The rates are shown below.

Table 7: Gas Rates – Feb 2020

Energy Charges	
Energy Charge	11.915 \$/GJ
Overrun Charge over 83 GJ/day	10 \$/GJ
Transmission Charge	
Supply Charge	3427.3008 \$/Month
Distribution Charge	
Demand First 50 GJ	2.05896993333333 \$/GJ/day
Demand Next 31 GJ	1.92794536666667 \$/GJ/day
Metering Charge	22.959016 \$/Day

4.3 Cost Saving Calculation Tariffs

The variable costs used for cost savings calculations in this report are shown below. The electricity cost used is the current average variable rate. The electrical demand cost is taken directly from the billing data. The costs are as follows:

Electricity:	13 c/kWh
Elec. Demand:	\$120 /kVA/annum
Gas:	\$11.9 /GJ

5 Description of Building Systems

5.1 Building Description

Four Seasons Hotel, Sydney is a 36 storey building.

Level 1 – 5 are known as the podium. Function rooms, kitchens, offices, pool and spa, laundry and other services are in these levels.

Levels 6-31 are typical hotel floors, including 16 premium suites and 4 deluxe suites per floor.

Level 32 contains several premier suites and deluxe suites as well as a club lounge.

Levels 33 has 13 premier suites and deluxe suites

Level 34 has 7 suites

Level 35 is the plant room, consisting of an external plant area for the cooling towers and distribution pipework as well as an internal plant area containing chiller plant.

Level 36 has 4 penthouse apartments

5.2 HVAC System

The hotel has a central system to provide heating and cooling to the building.

5.2.1 Chiller Plant

The system has a chiller plant as shown in the figure below that supply chilled water to

- 5 AHUs (fresh outside air conditioning) located at Level 35 and FCUs located Level 1-5 via chilled water distribution pumps DChWP1 and DChWP2.
- Room FCUs via secondary chilled water distribution pumps SChWP1, SChWP2 and SChWP3.

The chilled water system primary side is separated by a hydraulic separator.

The chilled water system has three chillers, two condensers and pumps with the following details:

L	Description	Tag	Make/Model	Capacity	kW
35	Chiller MAPS (1 off)	Chiller 1	PowerPax/W096-2F-N-SD	800 kW	
35	Chillers ENGIE	Chiller 2	Quantum/W110-E2G-LL special	900 kW	
35	Chillers ENGIE	Chiller 3	Quantum/W110-E2G-LL special	900 kW	
35	Chilled water Primary Pumps MAPS (1 off)	CH. W 1	TECO ASHBYC-DCC 22kW		22.00
35	Chilled water Primary Pumps ENGIE	CH. W 2	GRUNDFOS/NBG125-100-200/195 A-E2-D-BQQE		5.50

L	Description	Tag	Make/Model	Capacity	kW
35	Chilled water Primary Pumps ENGIE	CH. W 3	GRUNDFOS/NBG125-100-200/195 A-E2-D-BQQE		5.50
35	Secondary Chilled water pumps (3 off)	Sec. CH. W 1,2,3			22.00
35	Cooling Towers	CT 1	Aqua-Cool/MSS187A	900 kW	
35	Cooling Towers	CT 2	Aqua-Cool/MSS187A	900 kW	
35	Condenser water pumps MAPS (1 off)	CWP 1	TECO ASFB-NEDC1 15kW		15.00
35	Condenser water pumps ENGIE (2 off)	CWP 2	GRUNDFOS/NBG150-125-315/275 A-E2-D-BQQE		18.50
35	Condenser water pumps ENGIE (2 off)	CWP 3	GRUNDFOS/NBG150-125-315/275 A-E2-D-BQQE		18.50
35	CW Heat Recovery-Pump		GRUNDFOS TP-100-200/2A-F-B-BQQE		5.50

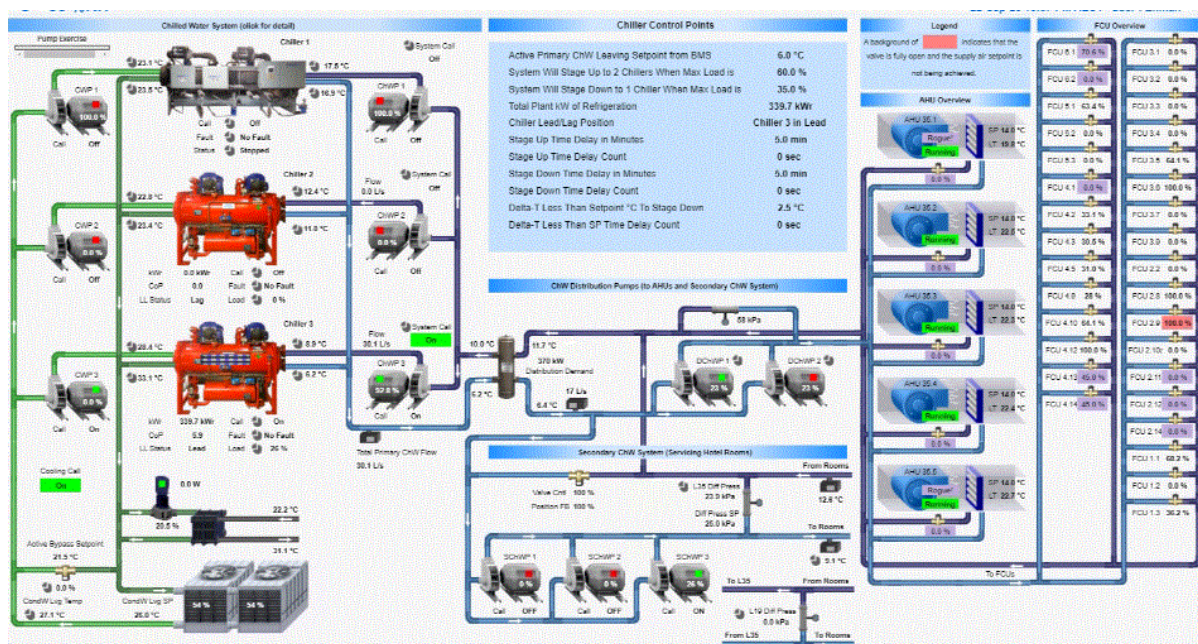


Figure 10: Service Area of Different HVAC System

5.3 Boiler Plant

There are two hot water boilers for domestic hot water and building heating as shown in the figure below.

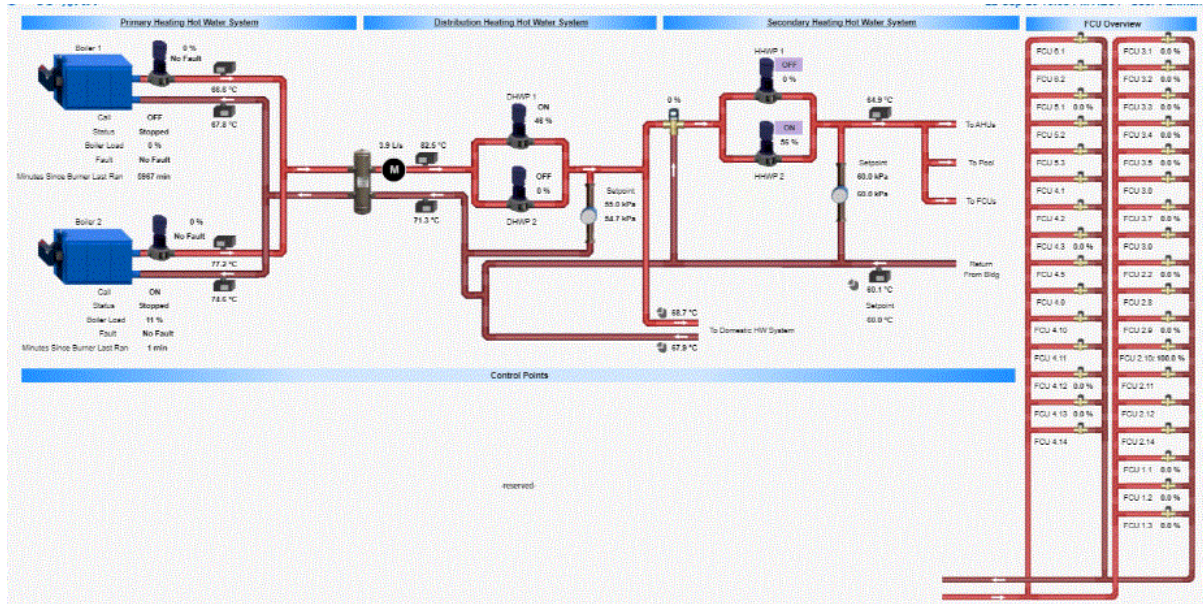


Figure 11: Hot Water Plant

The domestic hot water is heated by the supplied hot water via two domestic hot water pumps DWP1 and DWP2 and stored in 9 calorifiers from which hot water is supplied to different levels of the hotel as shown in the figure below.

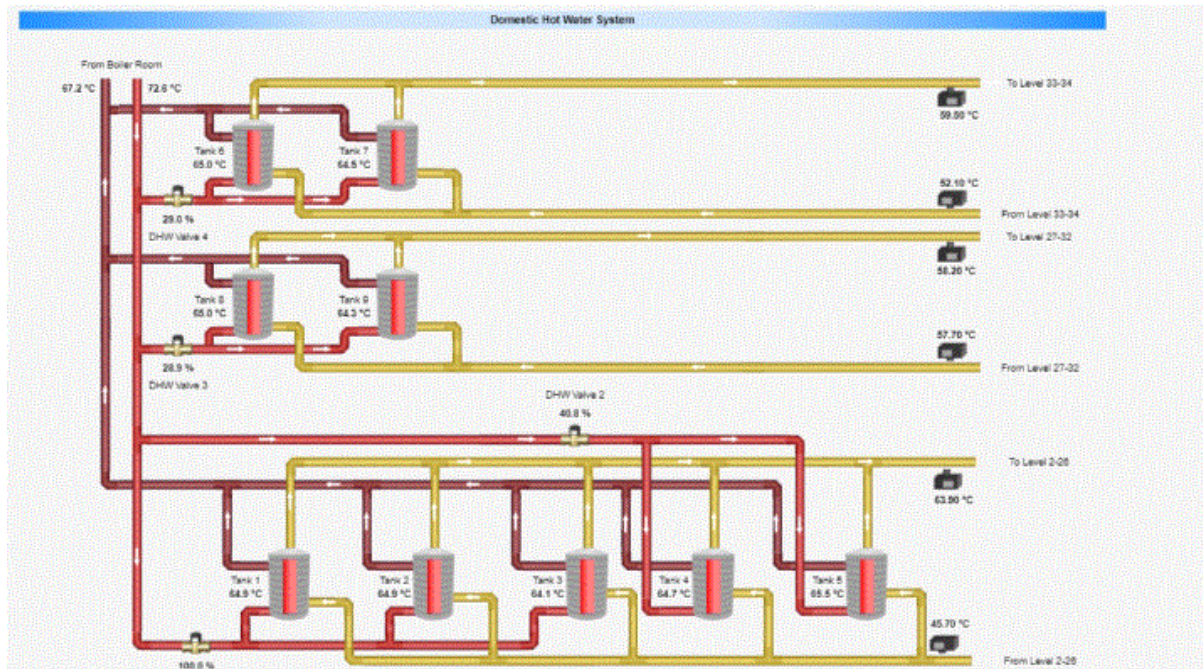


Figure 12: Domestic Hot Water System

The building heating and pool/spa heating hot water are supplied via two heating hot water pumps tagged as HHWP1 and HHWP2.

The details of the two hot water boilers are as follows:

Table 8: Hot Water Boilers

L	Location	Unit ID	Make/Model	Capacity/Output, kW
35	Hot Water Boilers (2 off)	Boiler 1,2	Rendamax/R3404	980kW

5.4 Ventilation

The ventilation fans comprise with supply and exhaust fans. A brief highlight of the fans is presented here:

- The car park has 2 large supply and exhaust fans. They have variable speed drives and their speed control based on CO monitor.
- Two large toilet exhaust fans with 9 kW fan motor for north and south sides have variable speed drives.
- Lobby/Corridor exhaust fan is large (9 kW fan motor) and has variable speed drive.

5.5 Building Envelop

The building has precast concrete wall and windows are double glazed.

5.6 Lighting

The lighting has a mix of LED, fluorescent and halogen lights. The breakdown of quantity and annual energy consumption shows that small quantities (11% of total counts) of fluoro lights contribute to over half of the lighting bill (53%).

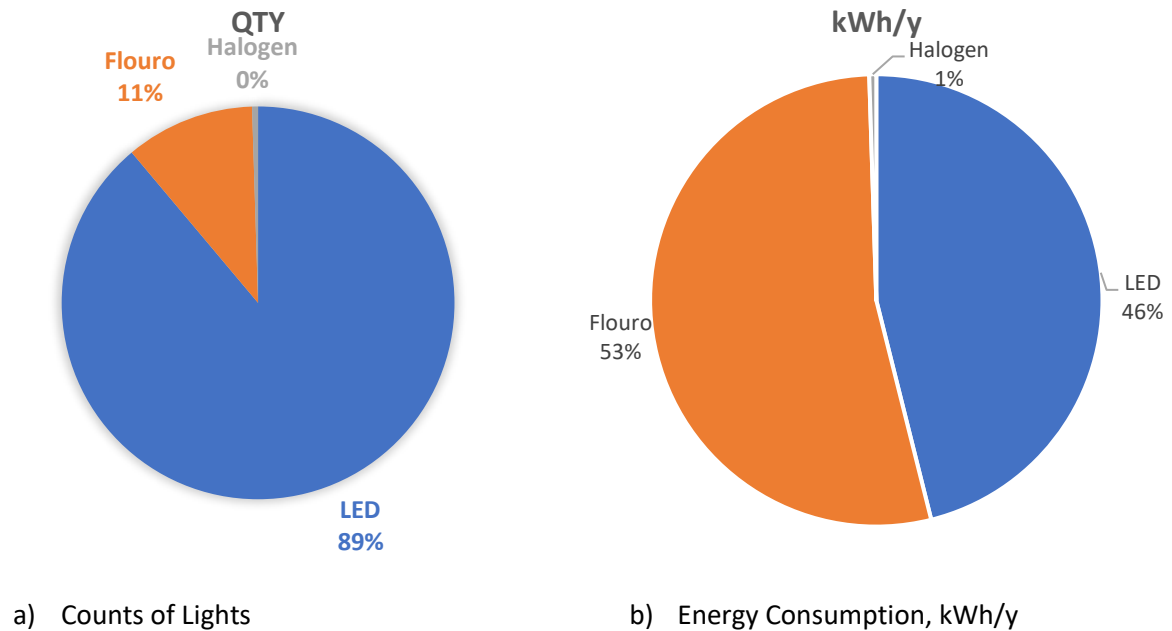


Figure 13: Lighting Distribution

5.7 Guest Rooms

The guest rooms have two pipes FCUs.

The hotel currently has a room management system to switches off the room lights and reset room temperature setpoints when unoccupied.

5.8 Equipment

The hotel has a variety of equipment including kitchen appliances, cool rooms, guest room appliances, and office equipment.

6 Energy Saving Opportunities

There are a number of energy saving opportunities which have been identified in this report.

6.1 Demand based Control of AHU Fan Speed

6.1.1 Description

There is an opportunity to implement demand-based fresh air supply by installing CO₂ monitors. Five AHUs supply fresh air to different areas of the hotel. They are:

Service Area	Unit ID	Rated Input (kW)
South Zone	AHU 35/1	18.50
Lobby & Corridor Air Handler	AHU 35/2	11.00
West Zone	AHU 35/3	15.00
North Zone	AHU 35/4	15.00
East Zone	AHU 35/5	11.00

The AHUs have VSDs but currently running at full speed most of the time. The summer and winter operations of AHU 1 is shown here. The hourly average data collected from BMS is used to generate trend plots.

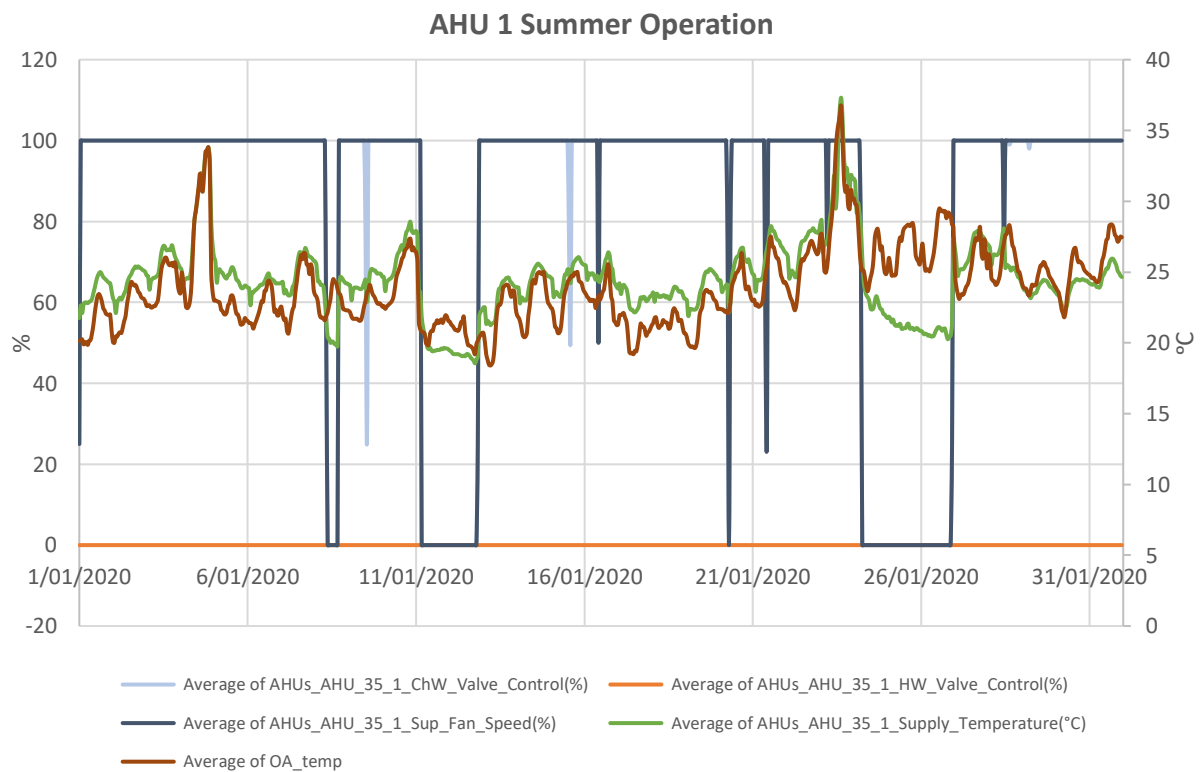


Figure 14: AHU 1 Summer Operation

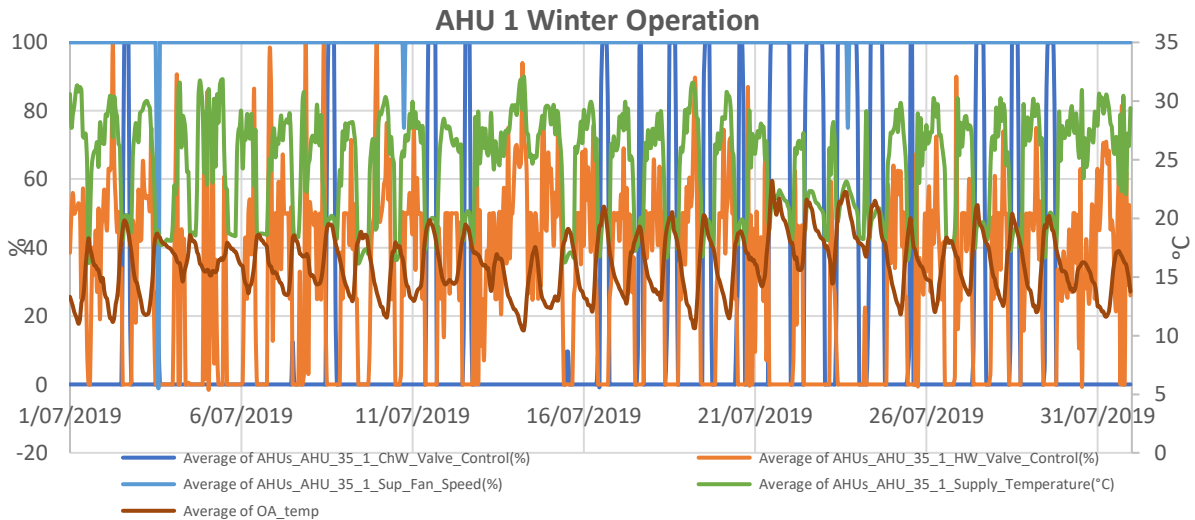


Figure 15: AHU 1 Winter Operation

The fan speed can be modulated based on demand measured by CO2 concentration of the areas. Currently, the control of chilled water and hot water valves is based temperature which can be enthalpy based.

6.1.2 Energy Savings

The energy savings will be achieved from the reduction in fan speed based on demand. The reduction of cooling and heating water from enthalpy-based control is not included here.

AHUs Present Energy Consumption	kWh/y	555,822
Estimated Savings from Speed Control	%	30%
Energy Savings	kWh/y	77,789
Cost Saving	\$/y	10,113

6.1.3 Project Cost

The project cost is estimated as follows:

	Description	Qty	\$/Unit	Cost \$	Sub-total \$
1	Equipment				
1.1	Controllers	3	\$1,100	\$3,300	
1.2	CO2 monitor	12	\$280	\$3,360	
					\$6,660
2	Engineering and Construction				
2.1	CO2 Sensor installation	12	\$500	\$6,000	
2.2	Controller HLI	3	\$500	\$1,500	
2.3	BMS Engineering and PM	1	\$10,000	\$10,000	
					\$17,500
	Total				\$24,160

Note:

- 12 CO2 sensors are allowed - 6 for lower 3 levels to control AHU 35/2 and 6 for upper levels. The CO2 is planned to implement on Exhaust Fans, e.g., toiler exhaust
- 3 controller are allowed-one for lower level and 2 for other levels. This will trade-off cost of wiring with controller

6.1.4 Project Summary

The summary of this project is shown in the table below.

Project Name	Electricity Saving (kWh/y)	Gas Saving (GJ/y)	Demand Saving (kVA)	Cost saving (\$/y)	Project cost (\$)	Simple Payback (y)	Estimated ESC Revenue	Revised Payback with ESC
Demand based Control of AHU Fan Speed	77,789	-	-	10,108	24,160	2.4	11,376	1.3

6.1.5 Recommendation and Other Benefits

The project is viable and recommended.

During the site visit, it was discussed that the AHUs are old and do not have return air connected with them. Four Seasons may consider an upgrade of the old AHUs with return air connections. The demand-based control will be more useful if that upgradation takes place in future and amount of return air can be controlled based on CO₂ concentration of serviced areas.

6.2 VSD on FCU Fans

6.2.1 Description

Four Seasons Hotel has installed variable speed drive to most of its fans that have over 2 kW fan motor size. There are a few fans left where VSD can be implemented and controlled based on demand measured by load (measured by valve positions) and schedule. The list of potential FCU fans are here:

L	Location	Unit ID	Rated Input (kW)
6	Laundry (south)	FCU 6/1	4.00
6	Laundry (North & Valet)	FCU 6/2	2.20
3	Gymnasium	FCU 4/11	4.00
2	Grain Bar	FCU 2/2	2.20

6.2.2 Energy Savings

Following energy savings are estimated for this project:

Current Energy Consumption by Fans	kWh/y	73,864
Estimated Savings		30%
Energy Savings	kWh/y	22,159
Cost Savings	\$/y	2,881

6.2.3 Project Cost

The project cost is estimated as follows:

	Description	Qty	\$/Unit	Cost \$	Sub-total \$
1	Equipment				
1.1	4 kW VSD	2	\$843	\$1,686	
1.2	2.2 kW VSD	2	\$728	\$1,456	
					\$3,142
2	Engineering and Construction				
2.1	VSD Installation	4	\$1,000	\$4,000	
2.2	BMS Software Upgrade	1	\$5,000	\$5,000	
					\$9,000
	Total				\$12,142

6.2.4 Project Summary

The summary of this project is shown in the table below.

Project Name	Electricity Saving (kWh/y)	Gas Saving (GJ/y)	Demand Saving (kVA)	Cost saving (\$/y)	Project cost (\$)	Simple Payback (y)	Estimated ESC Revenue	Revised Payback with ESC
VSD on FCU Fans	22,159	-	-	2,879	12,142	4.2	3,232	3.1

6.2.5 Recommendation and Other Benefits

The project is economically viable and recommended.

VSD implementation will increase low speed operation of fans which will reduce wear and tear of the fans. It is expected that it will increase the service life of the fans and reduce maintenance.

6.3 Demand Based FCU Fan Speed Control

6.3.1 Description

Four Seasons Hotel has installed variable speed drive to some FCU fans, but the speed of these fans does not modulate. The operation of the fans is observed from BMS and found that:

FCU	Speed
FCU 10a, FCU 2-8, FCU 2-9, FCU 3-5, FCU 4-2, FCU 4-5	100%
FCU 4-1	80%-100%

It is recommended to vary the speed of the FCU fans based on load which can be measured by valve positions-chilled water valve in cooling mode and hot water valve in heating mode.

6.3.2 Energy Savings

Following energy savings are estimated for this project:

Current Energy Consumption by the FCU Fans	kWh/y	158,468
Estimated Savings		28%
Energy Savings	kWh/y	44,371
Cost Savings	\$/y	5,766

6.3.3 Project Cost

The estimated project cost is 8,000 for BMS programming upgrade and tuning.

6.3.4 Project Summary

The summary of this project is shown in the table below.

Project Name	Electricity Saving (kWh/y)	Gas Saving (GJ/y)	Demand Saving (kVA)	Cost saving (\$/y)	Project cost (\$)	Simple Payback (y)	Estimated ESC Revenue	Revised Payback with ESC
Demand Based FCU Fan Speed Control	44,371	-	-	5,766	8,000	1.4	6,480	0.3

6.3.5 Recommendation and Other Benefits

The project is economically attractive and recommended.

VSD implementation will increase low speed operation of fans which will reduce wear and tear of the fans. It is expected that it will increase the service life of the fans and reduce maintenance.

6.4 Car Park Fan Optimal Control

6.4.1 Description

There are two supply and return fans ventilating the carpark. The existing control of the fans is based on CO level as follows:

- Minimum fan speed of 50% at ppm 5 or lower
- Max speed capped at 90% and
- Proportionally modulates between 50% to 90% fan speed for 5 to 15 ppm.

This control strategy produces sub-optimal operation of fans. The operation of the fan for a typical day is shown in the below figure. It shows that during the night-time, the CO level stays below 5 ppm

at 2.5 ppm (Approx.) from 9:45 PM to 6:45 PM. Further reduction of fan speed is possible due to minimum CO generation during this time.

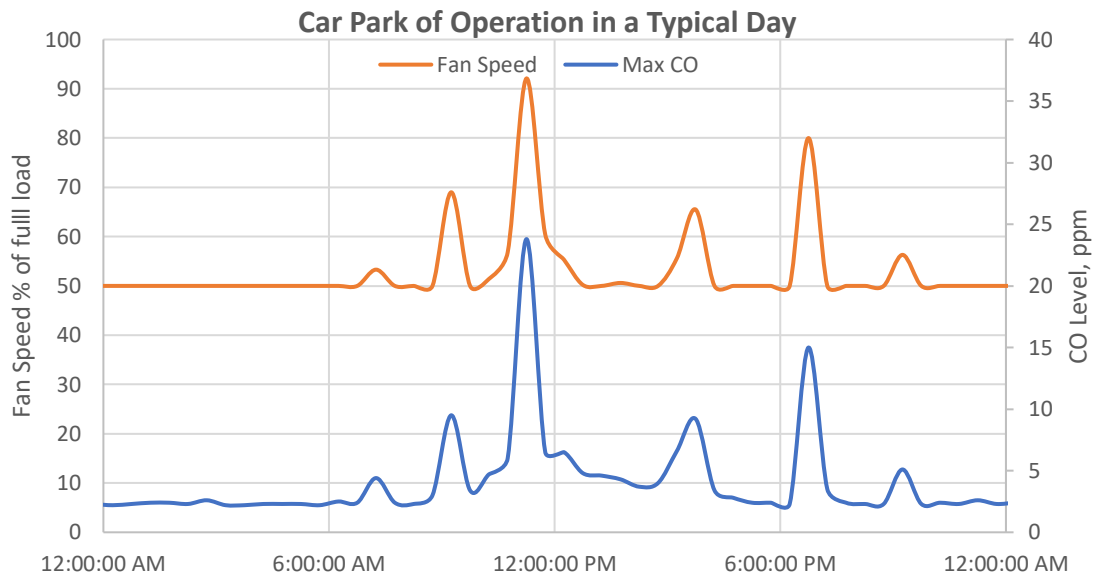


Figure 16: Carpark Fan Operation in a Typical Day

6.4.2 Energy Saving

The energy-saving is estimated based on the following control strategy:

- The minimum fan speed of 40% at ppm 5 or lower
- The max speed capped at 90% and
- Proportionally modulates between 40% to 90% fan speed for 5 to 15 ppm.

Note: The CO generation rate at the carpark is not considered for this estimation.

Location	Rated Input (kW)	LF	UF	Energy kWh/y	LF	UF	Energy kWh/y	Savings kWh/y
Carpark Supply Fan	15.60	0.14	1.0	19,728	0.09	1.0	11,647	8,080
Carpark Exhaust Fan	19.80	0.14	1.0	25,039	0.09	1.0	14,783	10,256
								18,336

6.4.3 Project Cost

The estimated project cost is \$4,000 approx. This includes the cost of re-programming BMS and subsequent tuning.

6.4.4 Project Summary

The summary of this project is shown in the table below.

Project Name	Electricity Saving (kWh/y)	Gas Saving (GJ/y)	Demand Saving (kVA)	Cost saving (\$/y)	Project cost (\$)	Simple Payback (y)	Estimated ESC Revenue	Revised Payback with ESC
Car Park Fan Optimal Control	18,336	-	-	2,383	4,000	1.7	2,672	0.6

6.4.5 Recommendation and Other Benefits

The project is economically attractive and recommended. The new control will allow the fans to run at much lower speed and subsequently, reduce wear and tear, increase life and reduce maintenance cost.

6.5 Refrigeration System Upgrade and Heat Recovery

6.5.1 Description

Four Seasons Hotel has 2 walk in freezers and 9 walk in cool rooms. The refrigeration cooling of these rooms is supplied by modular condensing units located in Level 2 Pool Plant Room. These condensing units are cooled by chilled water produced by central chiller plant. Due to low temperature of condensing water (chilled water temperature varies 6°C-10°C), the performance of the refrigeration system is good, but it calls for chillers to run throughout the winter.

The condensing units are running on hydrofluorocarbons (HFC) or hydrochlorofluorocarbon (HCFC) refrigerants-R404A, R409A and R134A. These refrigerants are under safety class A1 and A2 which indicate low toxic and low flammable. However, they have the following problems

- The higher performance of the refrigeration system is achieved with the expense of energy to produce chilled water by the central chiller plant
- Operation of central chiller plant throughout the winter
- Wastage of potential heat from the condenser
- All these refrigerants have high global warming potential (GWP)

There is a growing interest in natural refrigerant like CO₂ which has great potential in heat recovery. CO₂ transcritical refrigeration system is energy efficient and high-quality heat can be recovered from refrigerant gas cooling. Due to a large and consistent domestic hot water (DHW) and building heating hot water demand in the hotels, the CO₂ refrigeration system has potential in the hotel.

6.5.2 Energy Savings

The energy savings estimate is based on

- Heat recovery from gas cooling which can replace existing water heating by gas or electricity. High-quality heat can be recovered which can be used for nearby DHW or steam boiler.
- Reduction of chiller energy since chilled water use will be discarded

- The electrical energy savings of compressors is not included

Refrigeration Cooling Capacity	kW	44.24
Compressor Motor Capacity	kW	11.03
Total Heat Rejection	kW	55.27
Average Load Factor		0.6
Heat Recovery Potential:		
Potential Heat Recovery	kWh/y	290,499
Savings: Lower Level Electric Heating	kWh/y	283,824
Savings: Surplus for Gas Heating	GJ/y	24
Chiller Energy Savings:		
Chiller IPLV		10.32
Savings: Electrical Energy Savings from Chiller	kWh/y	46,915

6.5.3 Project Cost

The project cost is estimated as follows:

	Description	Qty	\$/Unit	Cost \$	Sub-total \$
1	Equipment				
1.1	TCO2 Rack	1	\$129,700	\$129,700	
1.2	Evaporator and Others	1	\$54,218	\$54,218	
					\$183,918
2	Engineering and Construction				
2.1	Mechanical Installation	1	\$50,000	\$50,000	
2.2	Engineering and PM	1	\$20,000	\$20,000	
					\$70,000
	Total				\$253,918

6.5.4 Project Summary

The summary of this project is shown in the table below.

Project Name	Electricity Saving (kWh/y)	Gas Saving (GJ/y)	Demand Saving (kVA)	Cost saving (\$/y)	Project cost (\$)	Simple Payback (y)	Estimated ESC Revenue	Revised Payback with ESC
Refrigeration System Upgrade and Heat Recovery	330,739	24	-	43,262	253,918	5.9	48,608	4.7

Note: An estimated 3,038 ESCs can be generated for the project which can be generate up to \$48,608 ESC revenue. This can lower the **payback to 4.7 years**.

6.5.5 Recommendation and Other Benefits

The proposed system has great potential and further investigation is recommended. The cost for further investigation can be subsidised by the City of Sydney 2nd year technical study grant or DPIE grant. The scopes of the assessment are as follows:

- Load calculation of freezer and cool rooms.
- Load profile study of the refrigeration system and hot water demand in lower levels
- System sizing calculation for quotation
- Business case preparation

If the project is proved economically attractive for the hotel based on the assessment and is approved for implementation, government grants can be looked for to subsidise the project implementation cost.

The benefits of the project are as follows:

- No more call for chiller operation throughout the winter
- Reduction of gas or electricity consumption for producing hot water
- Potential reduction of energy consumption from the increased efficiency
- Increased safety and reduced cost since CO2 is safe and inexpensive refrigerant
- Decreased environmental impact and future proof technology

6.6 Lighting Upgrade

6.6.1 Description

The hotel has fluorescent tubes in following areas.

L	Description	QTY
35	Plantroom	55
34	Corridor	4
33	Corridor	8
32	Club lounge	8
6-31	Service area	56
5	Housekeeping	164
4	Staff amenities	114
3	Admin offices	73
1	BOH offices	46
1	Front drive	8
B	Carpark/firestairs	489
FS	Main tower	165

The tubes are 36 W each with ballast of approx. 8 W. Energy can be saved by upgrading these lights to LED tubes of 18 W with no ballast.

The hotel has dichroic light in the following area:

L	Description	QTY
1	Mode restaurant	55

These dichroic lights have 35 watt each and can be upgraded with LED Downlight of 10W 800 LM.

6.6.2 Energy Savings

The energy savings will be coming from the lower wattage of the replacement lights. The existing energy consumption of these lights was estimated based on usage and wattage of the lights and the savings is determined by deducting energy consumption of LED lights from the existing energy consumption. The same usage was assumed for both LED and existing energy consumption. The summary calculation of savings is shown in the table below:

L	Description	QTY	Rated Input (W)	Ballast Input (W)	UF	Energy, kWh/y	NEW Rated Input (W)	Energy, kWh/y	Savings, kWh/y	Cost Savings, \$/y
35	Plantroom	55	36	8	1.00	21,199	18	8,672	12,527	1,628
34	Corridor	4	36	8	1.00	1,542	18	631	911	118
33	Corridor	8	36	8	1.00	3,084	18	1,261	1,822	237
32	Club lounge	8	36	8	0.75	2,313	18	946	1,367	178
6-31	Service area	56	36	8	1.00	21,585	18	8,830	12,755	1,658
5	Housekeeping	164	36	8	0.50	31,606	18	12,930	18,676	2,428
4	Staff amenities	114	36	8	0.50	21,970	18	8,988	12,982	1,688
3	Admin offices	73	36	8	0.50	14,069	18	5,755	8,313	1,081
1	BOH offices	46	36	8	0.50	8,865	18	3,627	5,238	681
1	Front drive	8	36	8	1.00	3,084	18	1,261	1,822	237
B	Carpark/firestairs	489	36	8	1.00	188,480	18	77,106	111,375	14,479
FS	Main tower	165	36	8	1.00	63,598	18	26,017	37,580	4,885
1	Mode restaurant	55	35		0.75	12,647	10	3,614	9,034	1,174
Total		1190				394,040		159,638	234,402	30,472

6.6.3 Project Cost

The project cost is \$132,040 based on cost estimate provided by a supplier. This price included LED tube and batten and the project will be eligible NSW energy savings certificates (ESCs). The estimated ESC revenue based on the energy savings is \$34,416.

6.6.4 Project Summary

The summary of this project is shown in the table below.

Project Name	Electricity Saving (kWh/y)	Gas Saving (GJ/y)	Demand Saving (kVA)	Cost saving (\$/y)	Project cost (\$)	Simple Payback (y)	Estimated ESC Revenue	Revised Payback with ESC
Lighting Upgrade	234,402	-	14	32,135	132,040	4.1	34,304	3.0

With estimated ESC revenue the payback period reduces **to 3 years**.

6.6.5 Recommendation and Other Benefits

The project is economically attractive and recommended. The other benefits of the projects are:

- Reduction of maintenance cost since LED lights have extended service life (over 50000 hrs)
- Reduction of cooling load (interactive effect to HVAC)

Note: A quotation received from a supplier is annexed with this report. The energy savings calculation and no of ESCs offered by the supplier is more than the estimate shown in previous section. This is beneficial for Four Seasons since they are receiving more subsidy from ESC revenue. In addition, consideration of cost savings from reduced maintenance and cooling load reduction will make the project more attractive.

6.7 Air Curtain on Two Main Entrance Doors

6.7.1 Description

Four Seasons Hotel Has three entrance and exit doors with 3 W x 2.5 H. Two of them are sliding doors which operate frequently. There is a considerable draft of air to the main entrance doors which can be reduced using air curtain.

Air curtain is a convenient draft reduction method and can

reduce the air infiltration and loss of conditioned air from the space. This can also improve comfort levels and save heating and cooling energy.

The air curtains may include electric heating and are controlled by the BMS system or independently to optimise their operation. The fan speed and heating will be independently adjustable with three settings: 'low', 'high' and 'off'.

6.7.2 Energy Savings

The energy saving is estimated considering that each door is open for 6 hrs a day in an average.

Cooling loss	kWh(T)	358,194
COP		8.0
Electrical Energy	kWh/y	44,774
Electricity Cost	\$/kWh	0.13
Electrical Energy Cost	\$/y	5,821

6.7.3 Project Cost

The estimated project cost for 2 doors is 20,000.

6.7.4 Project Summary

The project summary is as follow:

Project Name	Electricity Saving (kWh/y)	Gas Saving (GJ/y)	Demand Saving (kVA)	Cost saving (\$/y)	GHG saving (tCO ₂ -e/y)	Project cost (\$)	Simple Payback (y)
Air Curtain on Two Main Entrance and Exit Doors	44,774	-	-	5,818	41	20,000	3.4

6.7.5 Recommendation and Other Benefit

The project is viable and recommended. This project will also improve the comfort level.

6.8 Upgrade of Old Thermal Storage Tanks

6.8.1 Description

Four Seasons hotel has five near 40 years old calorifiers (thermal storage of heat) as shown in the figure below.



Figure 17: Old Thermal Storage Tanks

The operation of the hot water system is observed for few times through BMS. The hot water valves for the old calorifiers were found wide open at 11:40 PM when the hot water valves for four other new thermal storage tanks were found near close. This indicates loss of thermal energy due to the poor performance of these tanks.

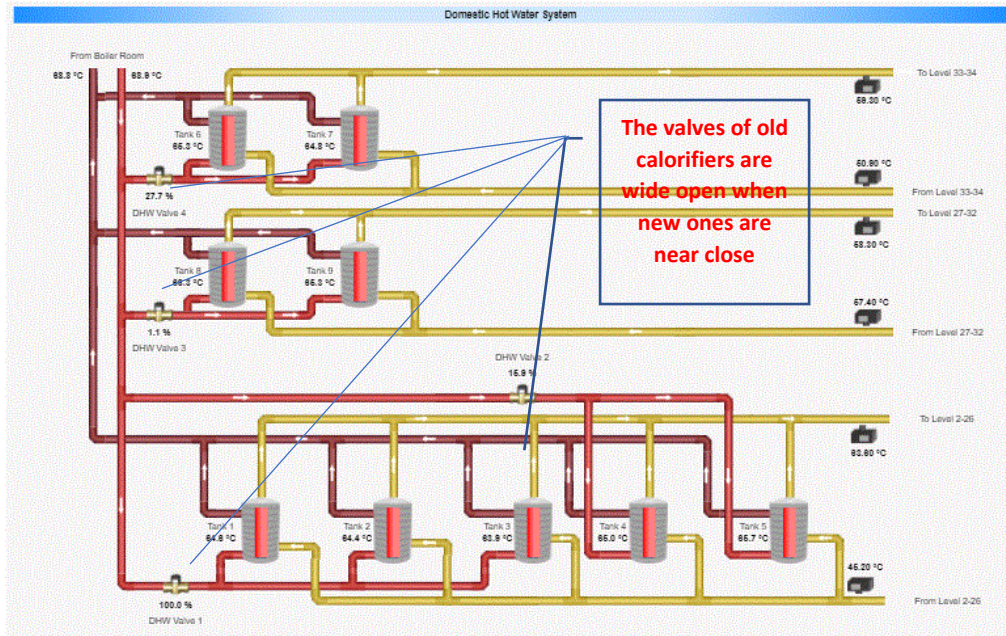


Figure 18: Calorifiers at 11:40 PM Nov 4, 2020

The efficiency of the storage tanks couldn't be measured due to lack of required data. Only historic data of the tank temperature is available on BMS and used to compare the performance of an old and a new tank and shown in the figure below which shows slow heat transfer from primary hot water to secondary DHW in the old calorifier. This comparison is indicative only since the flowrates of primary and secondary sides are unknown.

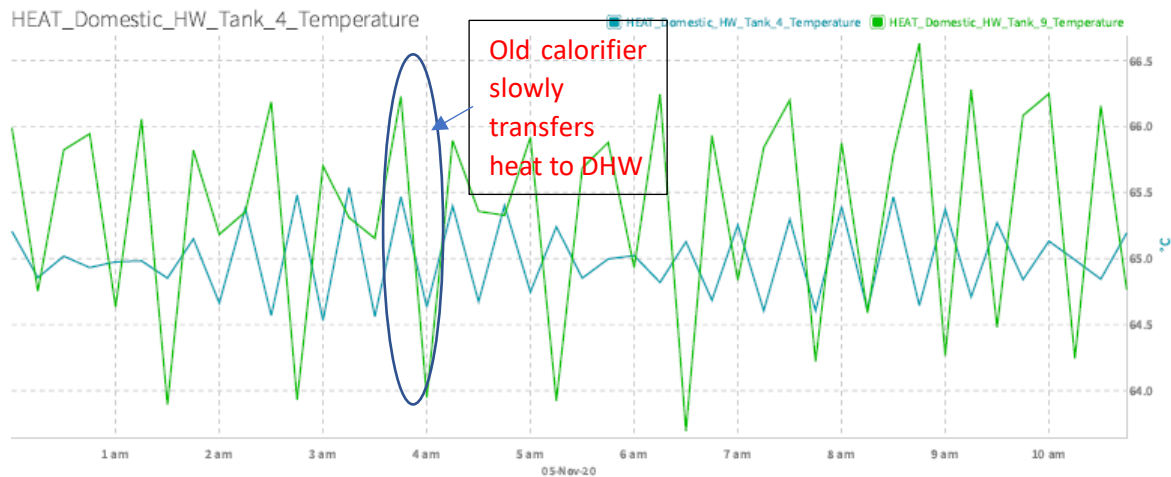


Figure 19: Comparison of old and new calorifiers heat transfer

6.8.2 Energy Savings

The energy savings will come from thermal energy loss from the calorifiers and reduction of electrical energy otherwise consumed by primary hot water circulation pumps. A 2.5% reduction of

boiler gas and 5% reduction of primary circulation pump electrical energy consumption is considered for the savings calculation. The mentioned savings are approximate.

Boiler Gas Consumption	GJ/y	11,941
Estimated Savings	%	2.5%
Energy Savings	GJ/y	299
Cost Saving	\$/y	3,553

Distribution HWP	kWh/y	11,936
Estimated Savings from Speed Control	%	5%
Energy Savings	kWh/y	597
Cost Saving	\$/y	78

6.8.3 Project Cost

The project cost is estimated as follows:

	Description	Qty	\$/Unit	Cost \$	Sub-total \$
1	Equipment				
1.1	Calorifiers	5	\$20,000	\$100,000	
1.2	Instrumentation	1	\$5,000	\$5,000	
					\$105,000
2	Engineering and Construction				
2.1	Mechanical Installation	1	\$40,000	\$40,000	
2.2	Electrical/Control Installation	1	\$10,000	\$10,000	
2.3	Engineering and PM	1	\$5,000	\$5,000	
					\$55,000
	Total				\$160,000

6.8.4 Project Summary

The summary of this project is shown in the table below.

Project Name	Electricity Saving (kWh/y)	Gas Saving (GJ/y)	Demand Saving (kVA)	Cost saving (\$/y)	GHG saving (tCO ₂ -e/y)	Project cost (\$)	Simple Payback (y)
Upgrade of Old Thermal Storage Tanks	597	299	-	3,635	20	160,000	44.0

6.8.5 Recommendation and Other Benefits

The project is not economically justified based on energy savings but can be considered since the calorifiers have finished their service life. Further investigation and engineering are recommended to take further action.

6.9 Upgrade of Booster Pumps

6.9.1 Description

There are two old booster pumps sets supplying cold water to Level 27-32 and Level 33-36.



a) Set 1 for L27-32



b) Set 2 for L33-36

Figure 20: Pressure Boosters

The booster pumps run ON/OFF and maintain a constant supply pressure using buffer tanks. During the site visit, it was observed that the supply pressure is high (1100 kPa).



Figure 21: High Booster Pump Supply Pressure

This operation is not energy efficient. The old sets can be replaced by new VSD based booster pump set which can better maintain the pressure with smooth operation and reduce of energy consumption.

6.9.2 Energy Savings

The energy savings will be achieved by the reduction of overall average speed of the pumps. The estimate is as follows:

Estimated Energy Consumption	kWh/y	118,260
Estimated Savings		20%
Energy Savings	kWh/y	23,652
Cost Savings	\$/y	3,073

6.9.3 Project Cost

The estimated project cost is \$71,044 for new system.

	Description	Qty	\$/Unit	Cost \$	Sub-total \$
1	Equipment				
1.1	Grundfos MPC-E 2 CRIE15-3	1	\$ 22,367	\$22,367	
1.2	Grundfos MPC-E 2 CRIE15-8	1	\$ 28,677	\$28,677	
					\$51,044
2	Engineering and Construction				
2.1	Mechanical Installation	1	\$20,000	\$15,000	
2.2	Engineering and PM	1	\$5,000	\$5,000	
					\$20,000
	Total				\$71,044

6.9.4 Project Summary

The summary of this project is shown in the table below.

Project Name	Electricity Saving (kWh/y)	Gas Saving (GJ/y)	Demand Saving (kVA)	Cost saving (\$/y)	GHG saving (tCO ₂ -e/y)	Project cost (\$)	Simple Payback (y)
Pressure Booster System Upgrade	23,652	-	-	3,073	22	71,044	23.1

6.9.5 Recommendation and Other Benefits

The project is not viable based on energy savings. However, Four Seasons Hotel may consider the project once the old systems end life.

6.10 Pipe Insulation of Hot Water System

6.10.1 Description

During the site visit, it was found that there are places where the hot water pipe does not have any insulation and/or the quality of the insulation is not good. This can cause heat loss from the piping.

6.10.2 Energy Savings

The amount of energy loss from the bare pipe is estimated which can be saved by insulating it. There are several places where there is no insulation as shown in the figure below:



Figure 22: Uninsulated Pipe in Hot Water Supply

The estimated length of bare pipe is 5 m of DN 50 copper pipe.

		Insulated Pipe	Bare Pipe	Savings
Surface Temperature	°C	26.5	70.0	
Heat Loss ⁴	W/m	14.6	226.4	211.8
Total Heat loss	W	72.8	1132.0	1059.2
Annual Savings	GJ/y			33.4

6.10.3 Project Cost

The project cost is estimated \$1,000.

6.10.4 Project Summary

The summary of this project is shown in the table below:

⁴ Estimated using engineering calculation method

Project Name	Electricity Saving (kWh/y)	Gas Saving (GJ/y)	Demand Saving (kVA)	Cost saving (\$/y)	GHG saving (tCO ₂ -e/y)	Project cost (\$)	Simple Payback (y)
Hot Water Pipe Insulation	-	33	-	398	2	1,000	2.5

6.10.5 Recommendation and Other Benefits/Considerations

This project is economically very attractive and recommended.

6.11 AHU Filter Pressure Drop

During the site visit, pressure drops against some AHU filters were found excessive. The cause can be dirt or poor quality. In the figure below, it is shown that AHU 3 and AHU 4 have high drop against the filter above and just below 50 Pa.



Figure 23: High Pressure Drop against AHU filter

6.11.1 Energy Savings

The amount of energy wasted due to lack of cleaning was estimated for AHUs and shown in the table below.

Table 9: Energy Wastage from AHU Filters

Estimated Average Air Flow Rate	m ³ /s	10
Estimated Average LF		0.95
Estimated Average UF		1
No AHUs with high pressure drop		2
Increased ΔP	Pa	30
% of time with high pressure drop		50%
Energy Wastage	kWh/y	2,497
Excess Cost	\$/y	325

6.11.2 Project Cost

There is no capital cost involved and the cleaning can be performed using operating expenditure.

6.11.3 Summary and Recommendation

It is recommended to check the units' performance more often and clean them early before they get dirty.

6.12 Equipment Replacement

There is an opportunity to save energy by replacing appliances with more energy efficient appliances. Replacing appliances for energy savings alone cannot generally be justified, however, it is recommended that when they do require replacing due to ageing, they should be replaced with equipment of a minimum of 5-star energy rating.

6.13 Maintenance and housekeeping

There is an opportunity to save energy by extending the site's activities from preventive or breakdown maintenance to an energy efficient maintenance program. Energy efficient maintenance can prove to be cheaper in the long term. Energy efficient maintenance can reduce energy costs and extend the life of the equipment. Some of these maintenance procedures are:

- Regular calibration check of thermostat control for the air conditioning system. If the controller is out of calibration the heating and cooling load may increase by as much as 20%.
- Cleaning the motor casings.
- Belt drive – V-belt is used for power transmission to fans in air handling/PACK units. If the tension of the belt is too high, too low or the belt is worn out, the transmission efficiency of the motor reduces. Transmission efficiency is dependent on pulley size, torque, under or over belting and V-belt construction. Belt tension is also an important parameter. Under and over belting can cause as high as a 5% efficiency drop.
- Rewinding of motors – It is recommended not to rewind motors more than two times. As motors fail, they are normally rewound. Rewinding a motor is cheaper than purchasing a new motor; however, rewinding reduces energy efficiency of the motor. The efficiency of a rewound motors drops due to the intense heating in stripping of old windings as well as using cheaper, thinner gauge wire. Rewinds are normally priced at 60-80% of the price of a standard new motor. On average, the efficiency of a motor decreases by about 1% each time the motor is rewound.
- Supply voltage – It is recommended to check the supply voltage to all distribution boards of the site. When the three-phase voltages are not equal, the motor losses increase substantially. This is normally due to unequal distribution of the single-phase loads, such as lighting. A modest phase imbalance of 2% can increase the motor losses by 25% or decrease overall motor efficiency by around 1.6%. When operating at less than 95% of design voltage,

motors typically lose efficiency by 2 to 4%. Supply voltage of over and under the rated voltage can also significantly reduce energy efficiency.

Appendix A: Project Summary with Energy Saving Certificates

Project Name	Electricity Saving (kWh/y)	Gas Saving (GJ/y)	Demand Saving (kVA)	Cost saving (\$/y)	GHG saving (tCO _{2-e} /y)	Project cost (\$)	Simple Payback (y)	Estimated ESC Revenue	Revised Payback with ESC
Demand based Control of AHU Fan Speed	77,789	-	-	10,108	72	24,160	2.4	11,376	1.3
VSD on FCU Fans	22,159	-	-	2,879	20	12,142	4.2	3,232	3.1
Demand Based FCU Fan Speed Control	44,371	-	-	5,766	41	8,000	1.4	6,480	0.3
Car Park Fan Optimal Control	18,336	-	-	2,383	17	4,000	1.7	2,672	0.6
Refrigeration System Upgrade and Heat Recovery	330,739	24	-	43,262	306	253,918	5.9	48,608	4.7
Lighting Upgrade	234,402	-	14	32,135	216	132,040	4.1	34,304	3.0
Air Curtains on Two Main Entrance and Exit Doors	44,774	-	-	5,818	41	20,000	3.4	6,544	2.3
Upgrade of Old Thermal Storage Tanks	597	299	-	3,635	20	160,000	44.0	2,688	43.3
Pressure Booster System Upgrade	23,652	-	-	3,073	22	71,044	23.1	3,456	22.0
Hot Water Pipe Insulation	-	33	-	398	2	1,000	2.5	288	1.8
TOTAL	796,820	356	14	109,456	756	686,304	9.3	119,648	8

Appendix B: Energy End-Use Breakdown Database

Table 10: Equipment

L	Location	Unit ID	Make/Model	Capacity	Rated Input (kW)	VSD	Qty	LF	UF	Energy kWh/y	Cost \$/y
38	Fire Stair 1 Pressurisation Fan	SF 38/1			4.85		1	0.8	-	-	-
37	Lift Motor Room	Guest 1-6	KONE/EP/23B/1		47.50		6	0.6	0.2	299,592	38,947
37	Lift Motor Room	Service 7-9	KONE/MX18G12		33.50		3	0.6	0.2	105,646	13,734
37	Lift Motor Room Exh	EF 38A/1			557.00		1	0.8	0	-	-
37	Lift Motor Room Exh	EF 38A/2			2.20		1	0.8	0	-	-
37	Lift Motor Room Exh	EF 38A/3			2.20		1	0.8	0	-	-
37	Lift Motor Room Exh	EF 38A/4			2.20		1	0.8	0	-	-
37	Lift Motor Room Exh	EF 38A/5			2.20		1	0.8	0	-	-
37	Lift Motor Room Exh	EF 38A/6			2.20		1	0.8	0	-	-
36	Fan Coil Units	ACU 36/1	Fujitsu		1.50		1	0.8	0.9	9,461	1,230
36	Fan Coil Units	ACU 36/2	Temperzone		0.73		1	0.8	0.5	2,558	333
36	Fan Coil Units	ACU 36/3	Temperzone		1.35		1	0.8	0.2	1,892	246
36	Fan Coil Units	ACU 36/4			1.35		1	0.8	-	-	-
36	Fan Coil Units	ACU 36/5			1.50		1	0.8	-	-	-
36	Fan Coil Units	ACU 36/6			0.73		1	0.8	-	-	-
36	Air Cooled Cond Unit for FCU	ACCU 38/1	Fujitsu		7.00		1	0.8	0.9	44,150	5,740
36	Air Cooled Cond Unit for FCU	ACCU 38/2	Temperzone		4.30		1	0.8	0.5	15,067	1,959
36	Air Cooled Cond Unit for FCU	ACCU 38/3	Temperzone		6.10		1	0.8	0.2	8,550	1,111
36	Air Cooled Cond Unit for FCU	ACCU 38/4			6.10		1	0.8	-	-	-
36	Air Cooled Cond Unit for FCU	ACCU 38/5			7.00		1	0.8	-	-	-
36	Air Cooled Cond Unit for FCU	ACCU 38/6			4.30		1	0.8	-	-	-
36	Lobby Fan Coil Unit	FCU 36/7			0.37		1	0.8	1.0	2,593	337
36	Fresh Air Supply Fans	SF 38/2			0.45		1	0.8	1.0	3,154	410
36	Fresh Air Supply Fans	SF 38/3			0.45		1	0.8	1.0	3,154	410
36	Toilet Exhaust Fans	EF 38/5			0.06		1	0.8	1.0	420	55
36	Toilet Exhaust Fans	EF 38/6			0.15		1	0.8	1.0	1,051	137

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L	Location	Unit ID	Make/Model	Capacity	Rated Input (kW)	VSD	Qty	LF	UF	Energy kWh/y	Cost \$/y
35	South Zone	AHU 35/1			18.50	1	1	0.9	1.0	145,854	18,961
35	Lobby & Corridor Air Handler	AHU 35/2			11.00	1	1	0.9	1.0	86,724	11,274
35	West Zone	AHU 35/3			15.00	1	1	0.9	1.0	118,260	15,374
35	North Zone	AHU 35/4			15.00	1	1	0.9	1.0	118,260	15,374
35	East Zone	AHU 35/5			11.00	1	1	0.9	1.0	86,724	11,274
35	Chiller MAPS (1 off)	Chiller 1	PowerPax/W096-2F-N-SD	800 kW		1	1			-	-
35	Chillers ENGIE	Chiller 2	Quantum/W110-E2G-LL special	900 kW		1	1			282, 241	36,691
35	Chillers ENGIE	Chiller 3	Quantum/W110-E2G-LL special	900 kW		1	1			495,452	64,409
35	Chilled water Primary Pumps MAPS (1 off)	CH. W 1	TECO ASHBYC-DCC 22kW		22.00	1	1			-	-
35	Chilled water Primary Pumps ENGIE	CH. W 2	GRUNDFOS/NBG125-100-200/195 A-E2-D-BQQE		5.50	1	1			18,621	2,421
35	Chilled water Primary Pumps ENGIE	CH. W 3	GRUNDFOS/NBG125-100-200/195 A-E2-D-BQQE		5.50	1	1			11,911	1,548
35	Secondary Chilled water pumps (3 off)	Sec. CH. W 1,2,3			22.00	3	3	0.3	0.5	86,724	11,274
35	Cooling Towers	CT 1	Aqua-Cool/MSS187A	900 kW		1	1			14,574	1,895
35	Cooling Towers	CT 2	Aqua-Cool/MSS187A	900 kW		1	1			11,859	1,542
35	Condenser water pumps MAPS (1 off)	CWP 1	TECO ASFB-NEDC1 15kW		15.00	1	1			-	-
35	Condenser water pumps ENGIE (2 off)	CWP 2	GRUNDFOS/NBG150-125-315/275 A-E2-D-BQQE		18.50	1	1			26,715	3,473
35	Condenser water pumps ENGIE (2 off)	CWP 3	GRUNDFOS/NBG150-125-315/275 A-E2-D-BQQE		18.50	1	1			28,252	3,673
35	CW Heat Recovery-Pump		GRUNDFOS TP-100-200/2A-F-B-BQQE		5.50	1	1			3,714	483

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L	Location	Unit ID	Make/Model	Capacity	Rated Input (kW)	VSD	Qty	LF	UF	Energy kWh/y	Cost \$/y
35	Hot Water Primary Pumps (2 off)	HWP 1,2	GRUNDFOSS/TP 08-120/2 AI-F-A-BUBE		1.50		2	0.9	0.6	13,919	1,809
35	Hot Water distribution pumps	DHWP 1	GRUNDFOSS TP 100-310/2A-F-B-BQQE		15.00	1	1			7,121	926
35	Hot Water distribution pumps	DHWP 2	GRUNDFOSS TP 100-310/2A-F-B-BQQE		15.00	1	1			4,815	626
35	Heating Hot Water Pumps	HHWP 1			5.50	1				-	-
35	Heating Hot Water Pumps	HHWP 2			5.50	1				13,227	1,720
35	Plantroom Exhaust	EF 38/2			1.50		1	0.8	1.0	10,512	1,367
35	Hydraulic Lift Motor room Exhaust	EF 35/2			0.18		1	0.8	1.0	1,261	164
35	Boiler Room Exhaust	EF 35/1			0.18		1	0.8	1.0	1,261	164
35	Fire Stair 2 Pressurisation Fan	SF 35/6			4.85		1	0	-	-	-
35	Booster Pump Set for L27-32	Set 1			11.00		2	0.9	0.5	86,724	11,274
35	Booster Pump Set for L33-36	Set 2			4.00		2	0.9	0.5	31,536	4,100
34	Toilet Exhaust North	EF 38/4			9.00	1	1	0.97	1.0	76,498	9,945
34	Toilet Exhaust South	EF 38/1			9.00	1	1	0.804	1.0	63,416	8,244
34	Lobby and Corridor Exhaust	EF 38/3			9.00	1	1	0.4	1.0	31,536	4,100
34	Living/Dining Room	FCU 34/1			0.37		1	0.8	0.5	1,296	169
34	Living/Dining Room	FCU 34/2			0.37		1	0.8	0.5	1,296	169
34	Living/Dining Room	FCU 34/3			0.37		1	0.8	0.5	1,296	169
34	Living/Dining Room	FCU 34/4			0.37		1	0.8	0.5	1,296	169
6	Laundry (south)	FCU 6/1			4.00		1	0.8	1.0	28,032	3,644
6	Laundry (North & Valet)	FCU 6/2			2.20		1	0.8	1.0	15,418	2,004
6	Boiler Room Supply	FVU 6/1			1.50		1	0.8	1.0	10,512	1,367
6	Main Laundry Exhaust	EF 6/21			1.72		1	0.8	1.0	12,054	1,567
6	Boiler & Valet Laundry Exhaust	EF 6/1			0.60		1	0.8	1.0	4,205	547
6	Fire Stairs 1A pressurisation	SF 6/1			4.85		1	0.8	-	-	-
6	Fire Stairs 3A pressurisation	SF 6/2			4.85		1	0.8	-	-	-
5	Staff Dining Room	FCU 5/1			1.50		1	0.8	0.5	5,256	683
5	Female Locker room A/C	FCU 5/3					1	0.8	0.5	-	-
5	Staff locker room Supply	FVU 5/1			0.75		1	0.8	0.5	2,628	342

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L	Location	Unit ID	Make/Model	Capacity	Rated Input (kW)	VSD	Qty	LF	UF	Energy kWh/y	Cost \$/y
5	Staff Toilet Exhaust	EF 5/1			1.10		1	0.8	1.0	7,709	1,002
5	Fire Stair 2A and 4A Pressurisation	SF 5/1			4.85		1	0.8	-	-	-
4	Administration	FCU 4/1			5.50	1	1	0.9	0.7	30,353	3,946
4	Security Personnel	FCU 4/3			0.37		1	0.8	0.8	2,074	270
4	Boutique	FCU 4/4			0.37		1	0.8	1.0	2,593	337
4	Special Functions	FCU 4/5			7.50	1	1	0.9	0.5	29,565	3,843
4	Health Club	FCU 4/6			0.75		1	0.8	0.7	3,679	478
4	Boutique (Lobby)	FCU 4/7			0.75		1	0.8	1.0	5,256	683
4	Boutique (Lobby)	FCU 4/8			0.75		1	0.8	1.0	5,256	683
4	Boutique (Lobby)	FCU 4/9			0.75		1	0.8	1.0	5,256	683
4	Gymnasium	FCU 4/12			1.50		1	0.8	1.0	10,512	1,367
4	Health Club Toilet Exhaust	EF 4/15			0.45		1	0.8	1.0	3,154	410
4	Supper Club Kitchen Exhaust Fan	EF 4/12			1.30		1	0.8	0.8	7,288	947
4	Hairdressers Exhaust	EF 4/16			0.45		1	0.8	1.0	3,154	410
4	Public Toilets	EF 5/1			1.10		1	0.8	1.0	7,709	1,002
4	Boutique Fresh Air Supply	SF 4/2			0.45		1	0.8	1.0	3,154	410
4	Fire Stair 6 Pressurisation	SF 4/1			4.85		1	0.8	-	-	-
4	Electric Water Heater	HWS 1-2	Rheem/61331508		14.40		2	0.4	1.0	100,915	13,119
4	Air Compressor Plant Room		ELGI/E11en-7	1.7 m3/min FAD	11.00		1	0.35	1.0	33,726	4,384
3	Gallery 1	FCU 3/1			0.75		1	0.8	0.7	3,679	478
3	Gallery 2	FCU 3/2			0.75		1	0.8	0.7	3,679	478
3	Gallery 3	FCU 3/3			0.75		1	0.8	0.7	3,679	478
3	Banquet office	FCU 3/4			0.37		1	0.8	0.7	1,815	236
3	Lobby and Restaurant South	FCU 3/5			3.00	1	1	0.9	1.0	23,652	3,075
3	Wine Store	FCU 3/6			0.75		1	0.8	0.8	4,205	547
3	Ballroom Sound Controls Room (East)	FCU 3/7			0.37		1	0.8	1.0	2,593	337
3	Wine Display Room	FCU 3/8			0.75		1	0.8	1.0	5,256	683
3	Wine store	FVU 3/1			0.75		1	0.8	1.0	5,256	683
3	Cooking Hood Exhaust Fan	EF 4/15			1.03		1	0.8	0.7	5,053	657
3	Cooking Hood Exhaust Fan	EF 4/16			1.36		1	0.8	0.7	6,672	867

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L	Location	Unit ID	Make/Model	Capacity	Rated Input (kW)	VSD	Qty	LF	UF	Energy kWh/y	Cost \$/y
3	Cooking Hood Exhaust Fan	EF 4/17			0.80		1	0.8	0.7	3,924	510
3	Dishwasher Exhaust	EF 4/11			0.45		1	0.8	0.7	2,208	287
3	Bakery Exhaust Hood	EF 4/10			0.77		1	0.8	0.7	3,777	491
3	Boundary Trap Exhaust Fan	EF 5/2			0.45		1	0.8	0.7	2,208	287
3	Can Wash Exhaust	EF 3/1			0.45		1	0.8	0.7	2,208	287
3	Bulk Store (East) Exhaust Fan	EF 3/5			0.18		1	0.8	1.0	1,261	164
3	Public Toilets	EF 5/1			1.10		1	0.8	1.0	7,709	1,002
3	Fire Stair 7 Pressurisation	EF 3/1			4.85		1	0.8	-	-	-
3	Gymnasium	FCU 4/11			4.00		1	0.8	0.7	19,622	2,551
2	Tenancy George street	FCU 2/1			1.50		1	0.8	0.8	8,410	1,093
2	Grain Bar	FCU 2/2			2.20		1	0.8	0.7	10,792	1,403
2	Brassiere	FCU 2/3			1.10		1	0.8	0.8	6,167	802
2	Boutique	FCU 2/4			0.75		1	0.8	0.7	3,679	478
2	Boutique	FCU 2/5			0.37		1	0.8	0.7	1,815	236
2	Boutique	FCU 2/6			0.37		1	0.8	0.7	1,815	236
2	Boutique	FCU 2/7			0.37		1	0.8	0.7	1,815	236
2	Lobby/Lounge	FCU 2/8			4.00	1	1	0.9	1.0	31,536	4,100
2	Admin	FCU 2/9			4.00	1	1	0.9	1.0	31,536	4,100
2	Club Bar	FCU 2/10			1.50	1	1	0.9	1.0	11,826	1,537
2	Functions Pantry	FCU 2/11			0.37		1	0.8	0.7	1,815	236
2	Functions Pantry	FCU 2/12			0.37		1	0.8	0.7	1,815	236
2	Computer Room	FCU 2/13			0.75		1	0.8	1.0	5,256	683
2	Service Corridor	FVU 2/1			1.10		1	0.8	1.0	7,709	1,002
2	Computer Room	ACCU 2/1			4.00		1	0.8	1.0	28,032	3,644
2	Computer Room	ACU 2/1			0.60		1	0.8	0.2	841	109
2	Tenancy George street	EF 2/			0.45		1	0.8	-	-	-
2	Tenancy George street	EF 2/1			0.45		1	0.8	-	-	-
2	Tenancy George street	EF 2/2			0.82		1	0.8	-	-	-
2	Tenancy George street	EF 2/3			0.60		1	0.8	-	-	-
2	Tenancy George street	EF 2/4			0.45		1	0.8	-	-	-

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L	Location	Unit ID	Make/Model	Capacity	Rated Input (kW)	VSD	Qty	LF	UF	Energy kWh/y	Cost \$/y
2	Dishwasher Exhaust	EF 3/2			0.18		1	0.8	1.0	1,261	164
2	Functions Pantry Roast Oven Exhaust	EF 3/3			0.77		1	0.8	0.8	4,317	561
2	Functions Pantry Cooking Hood Exhaust	EF 3/4			0.82		1	0.8	0.8	4,597	598
2	Brassiere Dishwasher Exhaust	EF 4/12			0.18		1	0.8	0.8	1,009	131
2	Brassiere Cooking Hood Exhaust	EF 4/13			1.72		1	0.8	0.8	9,643	1,254
2	Prefunction Toilet Exhaust	EF 4/14			0.45		1	0.8	1.0	3,154	410
2	Ballroom 1 and 2	FCU 4/2			5.50	1	1	0.593	0.7	19,990	2,599
2	Studios	FCU 4/10			1.50		1	0.8	0.7	7,358	957
2	Meat coolroom level 2		Bitzer/2FC-3.2y-40S	3.91	0.93		1	0.6	1.0	4,888	635
2	Garde Manger coolroom level 2		Bitzer/2DC-3.2y-40S	5.26	1.19		1	0.6	1.0	6,255	813
2	Pastry Coolroom level 2		Bitzer/2FC-3.2y-40S	3.91	0.93		1	0.6	1.0	4,888	635
2	Pastry Freezer level 2		Bitzer/4FC-3.2y-40S	3.26	1.10		1	0.6	1.0	5,782	752
2	Fruit and Veg coolroom 2 level 2		Bitzer/2EC-2.2y-40S	4.41	0.96		1	0.6	1.0	5,046	656
2	Dairy Coolroom level 2		Bitzer/2FC-3.2y-40S	3.91	0.93		1	0.6	1.0	4,888	635
2	Main Store Freezer level 2		Bitzer/2DES-3.2y-40S	4.39	1.41		1	0.6	1.0	7,411	963
2	Fruit and Veg coolroom 1 level 2		Bitzer/2EC-2.2y-40S	4.41	0.96		1	0.6	1.0	5,046	656
2	Beer Coolroom level 2		Bitzer/2GC-2.2y-40S	2.96	0.76		1	0.6	1.0	3,995	519
2	Fruit and Veg coolroom 3 level 2		Bitzer/2FC-3.2y-40S	3.91	0.93		1	0.6	1.0	4,888	635
2	Cooks Coolroom		Bitzer/2FC-3.2y-40S	3.91	0.93		1	0.6	1.0	4,888	635
2	Electric Water Heater	HWS 5-7	Rheem/61331508		14.40		3	0.45	1.0	170,294	22,138
2	Electric Water Heater	HWS	Rheem/401250		10.00		1	0.45	1.0	39,420	5,125
1	Engn Office 1	ACU 1/2			3.50		1	0.8	0.2	4,906	638
1	Engn Office 2	ACCU 1/2			0.37		1	0.8	1.0	2,593	337
1	Maintenance Workshop 1	ACU 1/1			3.50		1	0.8	1.0	24,528	3,189
1	Maintenance Workshop 2	ACCU 1/1			0.37		1	0.8	1.0	2,593	337
1	Carpark Supply Fan	SF 3/2			15.60	1	1	0.14	1.0	19,132	2,487
1	Fire Control Room Supply Fan	SF 1/1			0.45		1	0.8	1.0	3,154	410
1	Carpark Exhaust Fan	EF 2/4			19.80	1	1	0.14	1.0	24,283	3,157
1	Grease Trap Exhaust	EF 1/1			1.50		1	0.8	1.0	10,512	1,367
B	Electric Water Heater	HWS 3-4	Rheem/61331508		14.40		2	0.45	1.0	113,530	14,759

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L	Location	Unit ID	Make/Model	Capacity	Rated Input (kW)	VSD	Qty	LF	UF	Energy kWh/y	Cost \$/y
Lvl 6-32	Room fan coil units	60Y Tempzone			0.6		681	0.55	0.4	787,454	102,369

Table 11: Lighting

L	Description	Type	Lamp Type	QTY	Hours Operate	Rated Input (W)	Ballast Input (W)	LF	UF	Annual Consumption	Cost (\$/yr)
36	Corridor	T1	LED	14	24	12		1.0	1.00	1,472	\$191
35	Plantroom	T3	Fluoro	55	24	36	8	1.0	1.00	21,199	\$2,756
34	Corridor	T2	LED	30	24	10		1.0	1.00	2,628	\$342
34	Corridor	T3	Fluoro	4	24	36	8	1.0	1.00	1,542	\$200
34	Corridor	T4	LED	20	24	8	0		1.00	0	\$0
34	Suite One bed	T2	LED	182	6	10		1.0	0.25	3,986	\$518
34	Suite One bed	T4	LED	52	6	8		1.0	0.25	911	\$118
34	Suite One bed	T6-Makeup	LED	13	6	5		1.0	0.25	142	\$19
34	Suite Two bed	T2	LED	21	6	10		1.0	0.25	460	\$60
34	Suite Two bed	T4	LED	6	6	8		1.0	0.25	105	\$14
34	Suite Two bed	T6-Makeup	LED	2	6	5		1.0	0.25	22	\$3
33	Corridor	T2	LED	24	24	10		1.0	1.00	2,102	\$273
33	Corridor	T3	Fluoro	8	24	36	8	1.0	1.00	3,084	\$401
33	Corridor	T4	LED	18	24	8		1.0	1.00	1,261	\$164
32	Club lounge	T2	LED	63	12	10		1.0	0.50	2,759	\$359
32	Club lounge	T3	Fluoro	8	18	36	8	1.0	0.75	2,313	\$301
32	Club lounge	T4	LED	22	18	8		1.0	0.75	1,156	\$150
32	Club lounge	T5-Panel	LED	2	18	27		1.0	0.75	355	\$46
32	Corridor	T2	LED	25	24	10		1.0	1.00	2,190	\$285
32	Standard room	T2	LED	56	6	10		1.0	0.25	1,226	\$159
32	Standard room	T4	LED	16	6	8		1.0	0.25	280	\$36

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L	Description	Type	Lamp Type	QTY	Hours Operate	Rated Input (W)	Ballast Input (W)	LF	UF	Annual Consumption	Cost (\$/yr)
32	Standard room	T6-Strip	LED	24	6	8		1.0	0.25	420	\$55
32	Standard room	T6-Makeup	LED	8	6	5		1.0	0.25	88	\$11
32	Corner room	T2	LED	45	6	10		1.0	0.25	986	\$128
32	Corner room	T4	LED	15	6	8		1.0	0.25	263	\$34
32	Corner room	T6-Strip	LED	12	6	8		1.0	0.25	210	\$27
32	Corner room	T6-Makeup	LED	3	6	5		1.0	0.25	33	\$4
6-31	Standard room	T2	LED	2849	6	10		1.0	0.25	62,393	\$8,111
6-31	Standard room	T4	LED	804	6	8		1.0	0.25	14,086	\$1,831
6-31	Standard room	T6-Strip	LED	1221	6	8		1.0	0.25	21,392	\$2,781
6-31	Standard room	T6-Makeup	LED	407	6	5		1.0	0.25	4,457	\$579
6-31	Corner room	T2	LED	1650	6	10		1.0	0.25	36,135	\$4,698
6-31	Corner room	T4	LED	550	6	8		1.0	0.25	9,636	\$1,253
6-31	Corner room	T6-Strip	LED	440	6	8		1.0	0.25	7,709	\$1,002
6-31	Corner room	T6-Makeup	LED	110	6	5		1.0	0.25	1,205	\$157
6-31	Service area	T3	Fluoro	56	24	36	8	1.0	1.00	21,585	\$2,806
5	Housekeeping	T3	Fluoro	164	12	36	8	1.0	0.50	31,606	\$4,109
4	Staff amenities	T2	LED	45	12	10		1.0	0.50	1,971	\$256
4	Staff amenities	T3	Fluoro	114	12	36	8	1.0	0.50	21,970	\$2,856
4	Staff amenities	T4	LED	10	12	8		1.0	0.50	350	\$46
3	Gym	T2	LED	50	18	10		1.0	0.75	3,285	\$427
3	Admin offices	T3	Fluoro	73	12	36	8	1.0	0.50	14,069	\$1,829
3	Lobby ceiling	T9-Down	LED	30	24	42		1.0	1.00	11,038	\$1,435
3	Public space	T1	LED	180	24	12		1.0	1.00	18,922	\$2,460
3	Public space	T2	LED	64	24	10		1.0	1.00	5,606	\$729
2	Studios	T2	LED	153	12	10		1.0	0.50	6,701	\$871
2	Galleries	T2	LED	111	12	10		1.0	0.50	4,862	\$632
2	Kitchens	T5-Panel	LED	113	18	27		1.0	0.75	20,045	\$2,606

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2	Resi suite	T2	LED	41	6	10		1.0	0.25	898	\$117
2	Resi suite	T4	LED	6	6	8		1.0	0.25	105	\$14
2	Resi suite	T6-Strip	LED	30	6	8		1.0	0.25	526	\$68
2	Public space	T1	LED	77	24	12		1.0	1.00	8,094	\$1,052
2	Public space	T2	LED	44	24	10		1.0	1.00	3,854	\$501
1	Ballroom	Ballroom	LED	1	6	11016		1.0	0.25	24,125	\$3,136
1	Mode restaurant	T2	LED	30	18	10		1.0	0.75	1,971	\$256
1	Mode restaurant	T4	LED	27	18	8		1.0	0.75	1,419	\$184
1	Mode restaurant	T6-Strip	LED	41	18	8		1.0	0.75	2,155	\$280
1	Mode restaurant	T7-dichroic	Halogen	55	18	37		1.0	0.75	13,370	\$1,738
1	BOH offices	T3	Fluoro	46	12	36	8	1.0	0.50	8,865	\$1,152
1	Front drive	T1	LED	10	24	12		1.0	1.00	1,051	\$137
1	Front drive	T2	LED	26	24	10		1.0	1.00	2,278	\$296
1	Front drive	T3	Fluoro	8	24	36	8	1.0	1.00	3,084	\$401
1	Lobby	T1	LED	68	24	12		1.0	1.00	7,148	\$929
1	Grain Bar	T2	LED	48	18	10		1.0	0.75	3,154	\$410
1	Grain Bar	T6-Strip	LED	35	18	8		1.0	0.75	1,840	\$239
B	Carpark/firestairs	T3	Fluoro	489	24	36	8	1.0	1.00	188,480	\$24,502
B	Entry ramp	T8-bay light	LED	2	24	80		1.0	1.00	1,402	\$182
B	Exit ramp	T8-bay light	LED	3	24	80		1.0	1.00	2,102	\$273
FS	Main tower	T3	Fluoro	165	24	36	8	1.0	1.00	63,598	\$8,268
Total										709,733	\$92,265

Table 12: Guest Room Appliances

Guest Room Appliances	Quantity	Rated Input (KW)	LF	UF	Current Annual Consumption (kWh/y)	Cost (\$/y)
T.V	531	0.1	1	0.13	58,145	\$7,559
Kettle	531	0.7	1	0.002	6,512	\$847
Coffee Machine	531	1.26	1	0.002	11,722	\$1,524
Hair Dryer	531	2	1	0.005	46,516	\$6,047
Iron	531	2.3	1	0.005	53,493	\$6,954
Fridge	531	0.05	0.5	1	116,289	\$15,118
Total					292,676	\$38,048

Table 13: Gas

L	Location	Unit ID	Make/Model	Capacity/Output, kW	Rated Input, kW	Plant type	Qty	LF	UF	Energy GJ/y
35	Hot Water Boilers	Boiler 1	Rendamax/R3404	980kW	1037.797	Hot Water Boiler	1	0.31	0.5	5,299
36	Hot Water Boilers	Boiler 2	Rendamax/R3404	980kW	1037.797	Hot Water Boiler	1	0.31	0.7	6,643
5	Laundry Boiler		Fulton/20J	250 kW	312.5	Steam Boiler	1	0.35	1.0	3,449

Appendix C: Quotes

Lighting Upgrade

Proposal Summary

Total Savings	
Current Annual Lighting Cost	\$50,559
LED Annual Running Cost	\$18,862
Total Annual Savings with LED	\$31,697
10 Year Savings *	\$312,040
10 Year Maintenance Savings	\$90,132
Return On Investment	36%
Payback Period	33 months
Energy Savings are based on a rate of \$0.31 / kWh *Includes inflation of 5% p.a on energy costs and CPI increase of 2% p.a on replacement costs	

Environmental Benefits

Annual Energy Saving (kWh)	214,793
No. of ESC's	252.1
Carbon Abatement (kg)	257,752

Project Investment

Total Project Value	\$132,040.00
Discount (Including ESC's)	\$45,184.55
Investment	\$86,855.45^f

^fProposal is valid for 21 days

